

Machine Learning

CMPSCI 589

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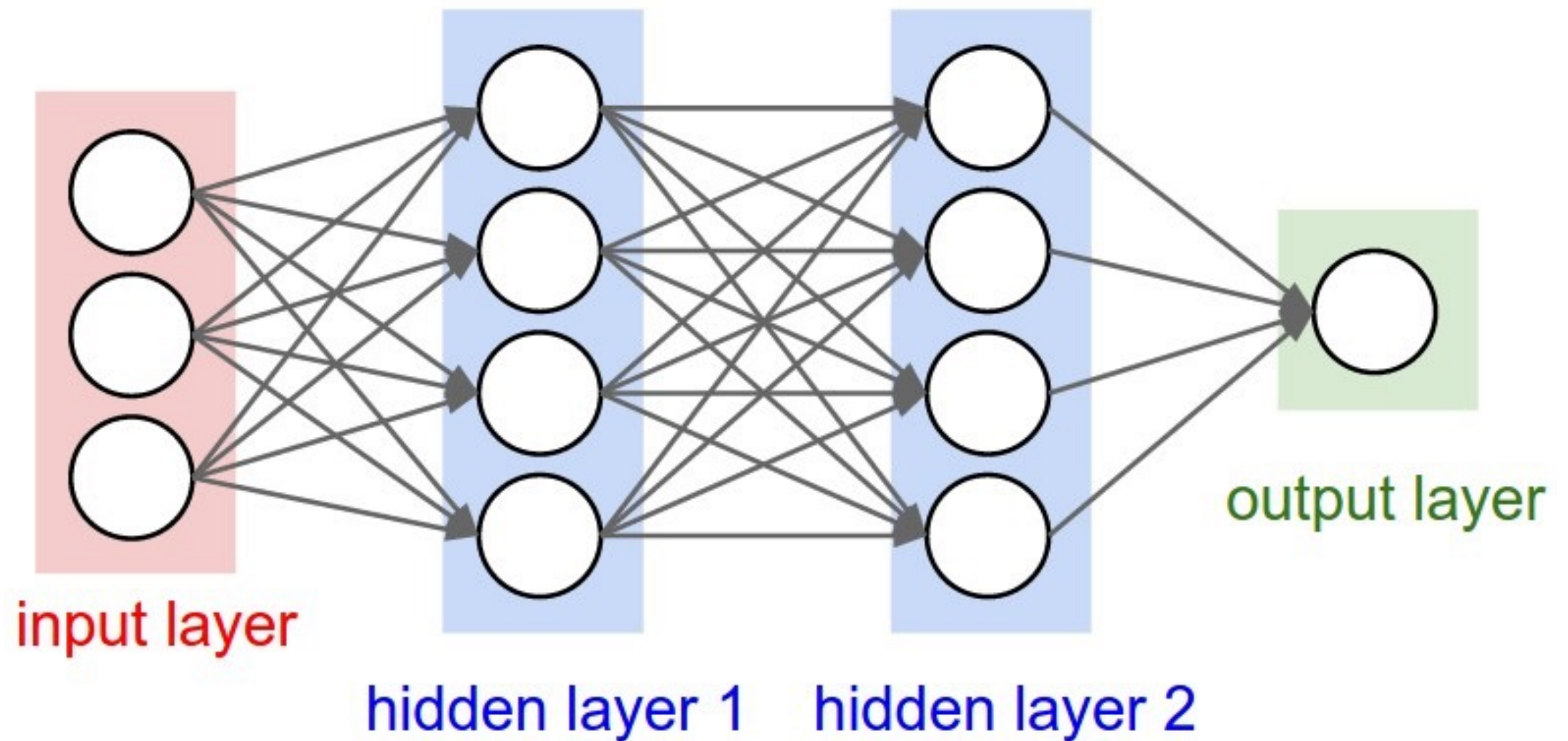
Neural Networks (pt. 3)

Vectorization

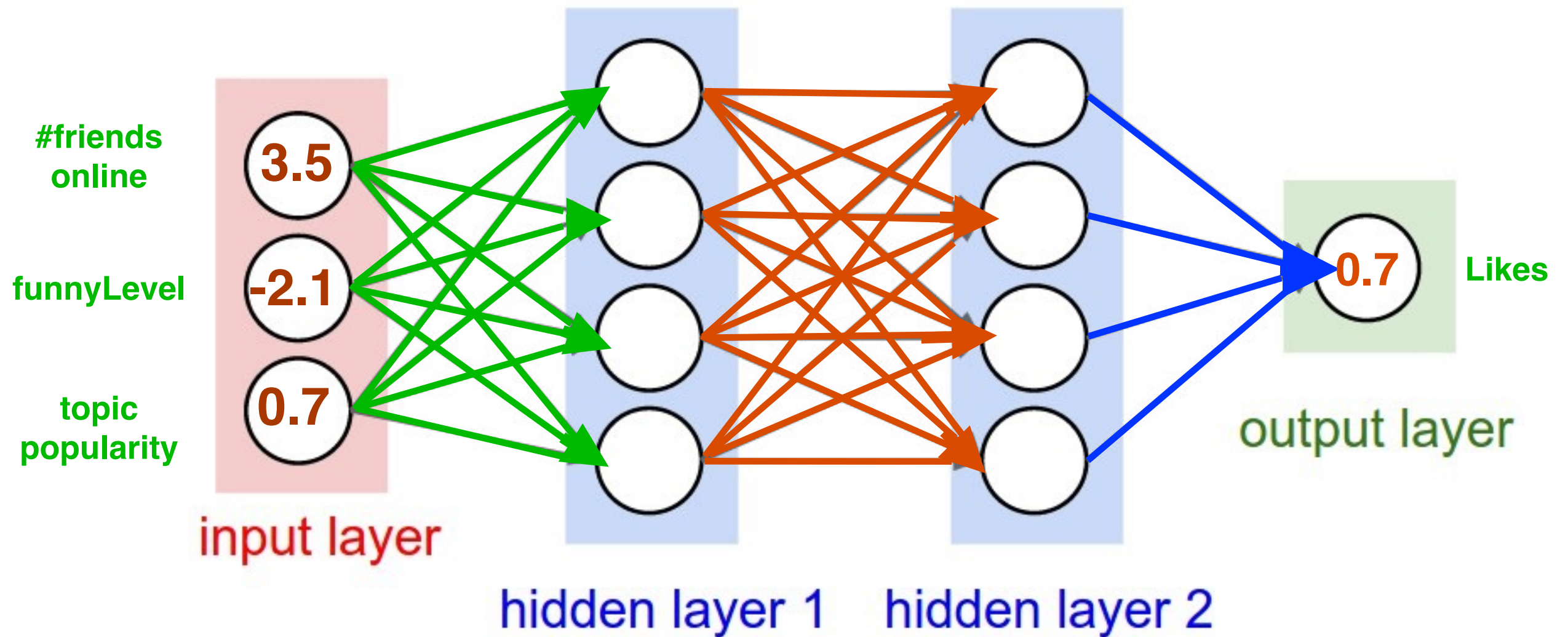


Vectorization

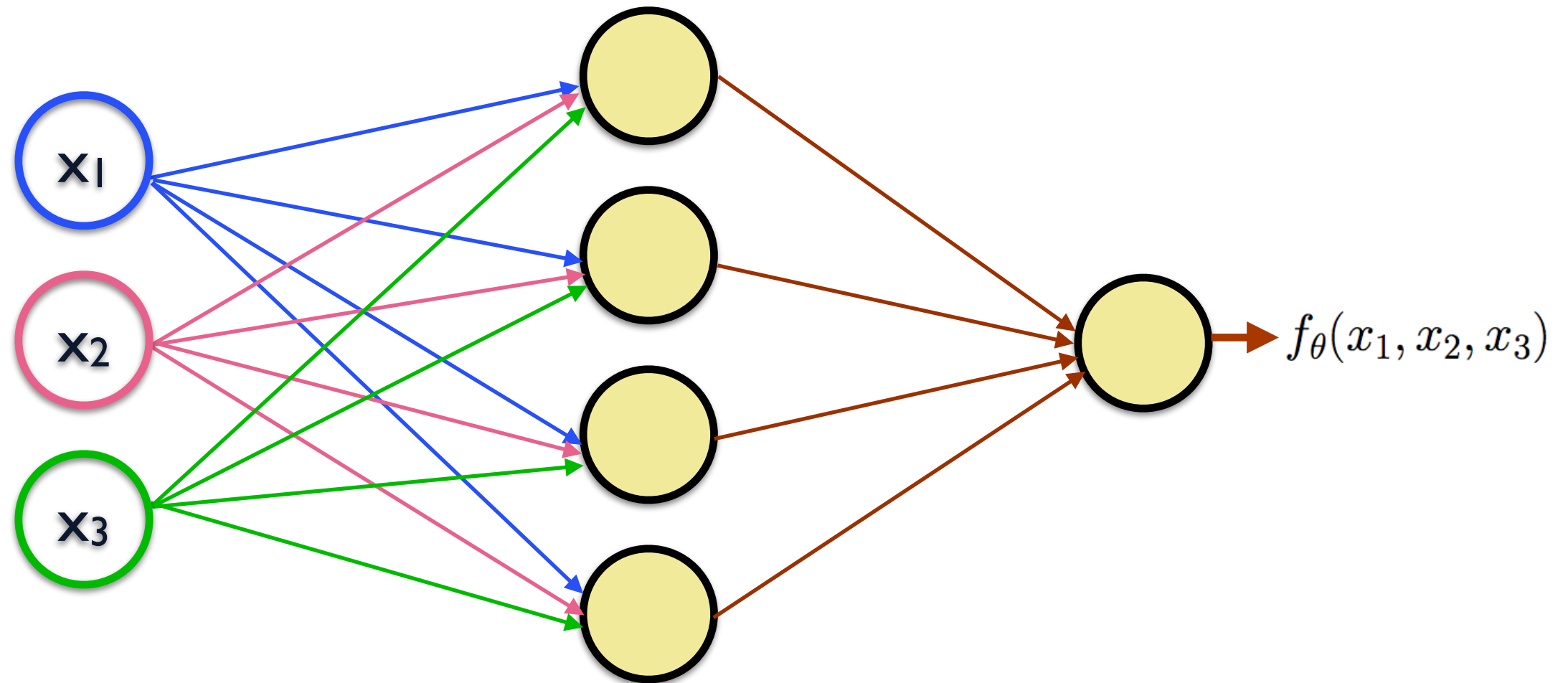
Multi-Layer Neural Networks



Multi-Layer Neural Networks



Multi-Layer Neural Networks

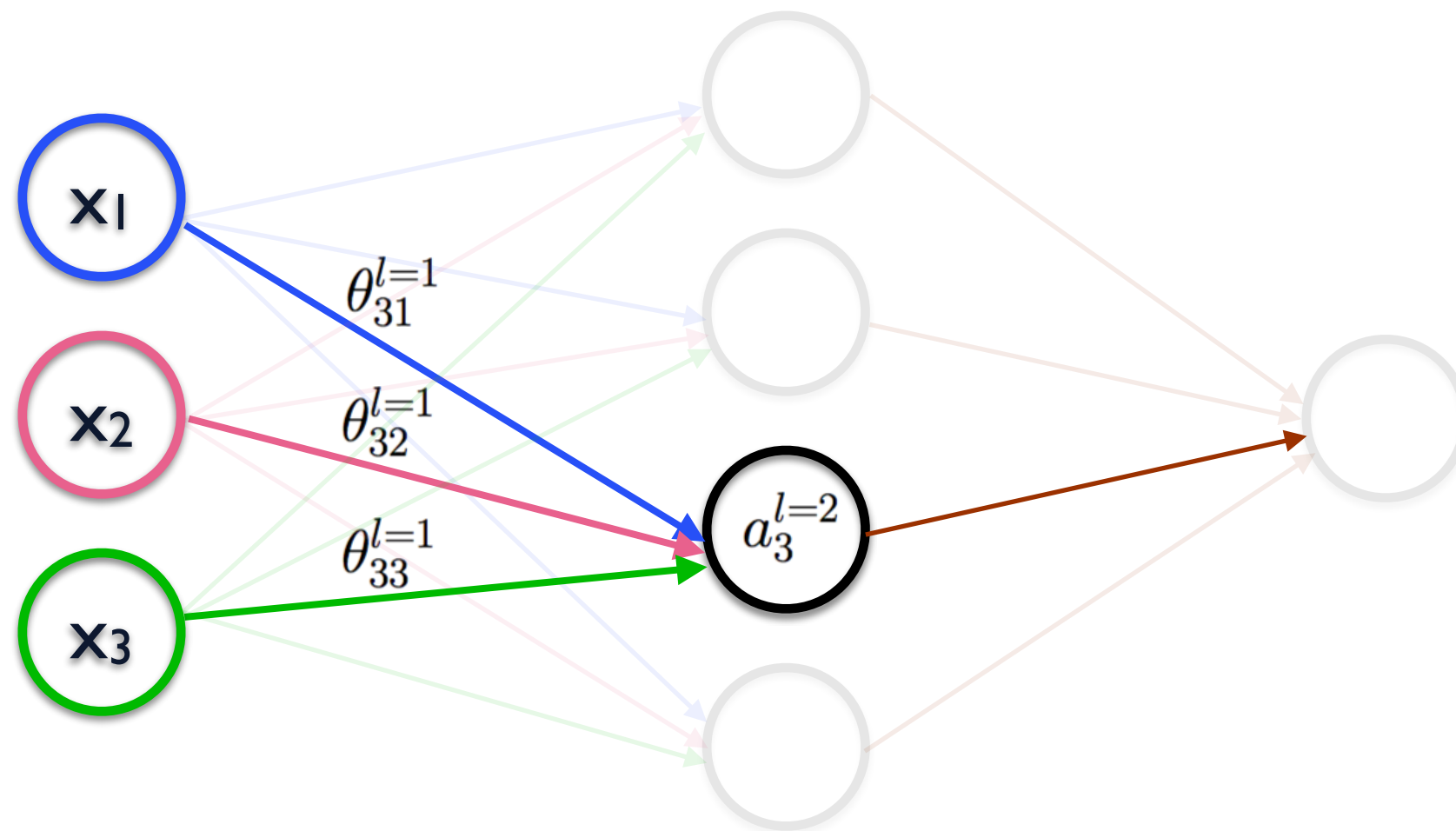


Network (graph) of connected neurons

Neural Networks

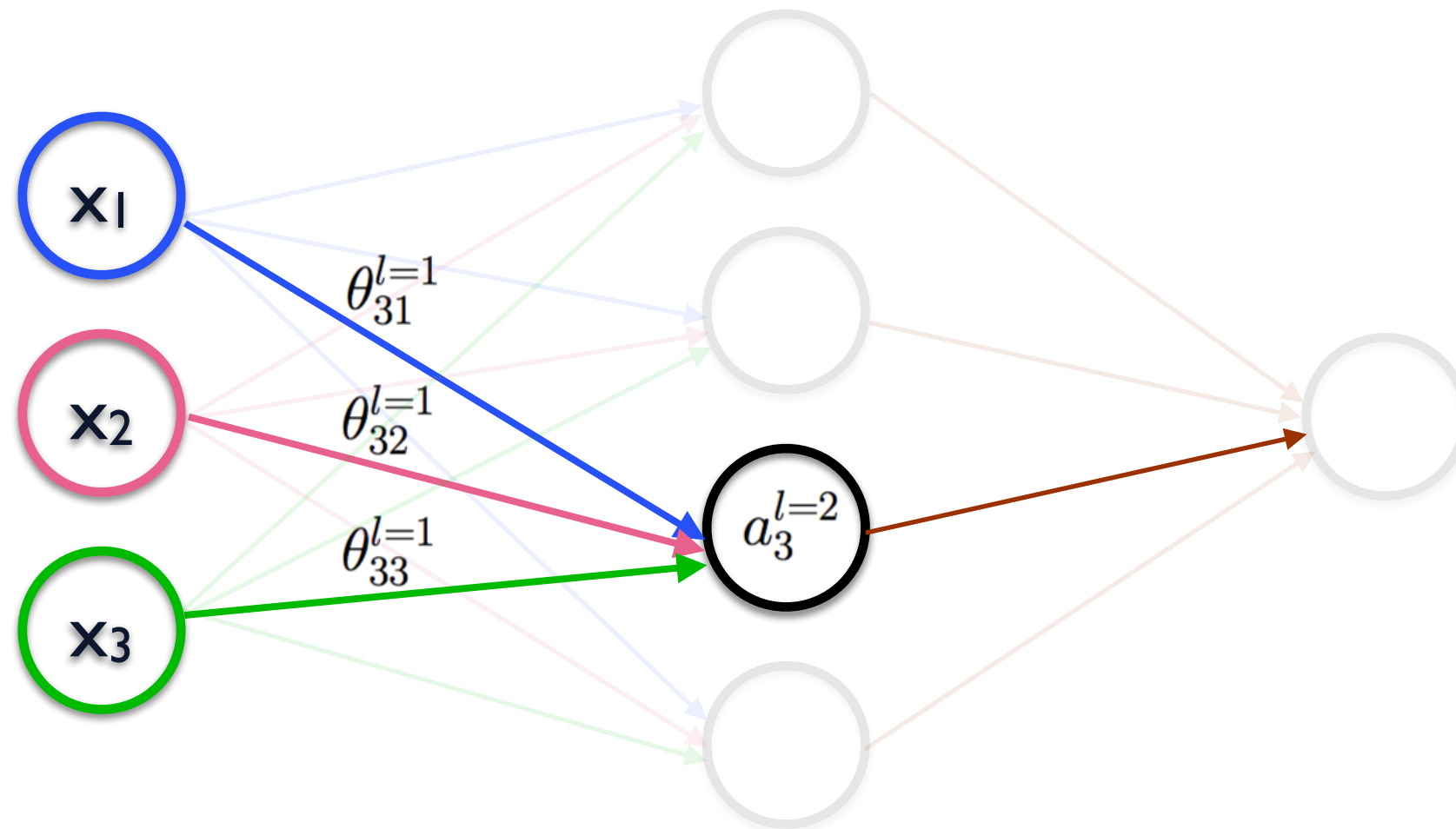
θ_{ij}^l : weight connecting the ***j*-th** neuron (in the ***l*-th** layer) to the ***i*-th** neuron in the next layer

a_j^l : output/activation of the ***j*-th** neuron in the ***l*-th** layer



Computing the output/activation of a single neuron

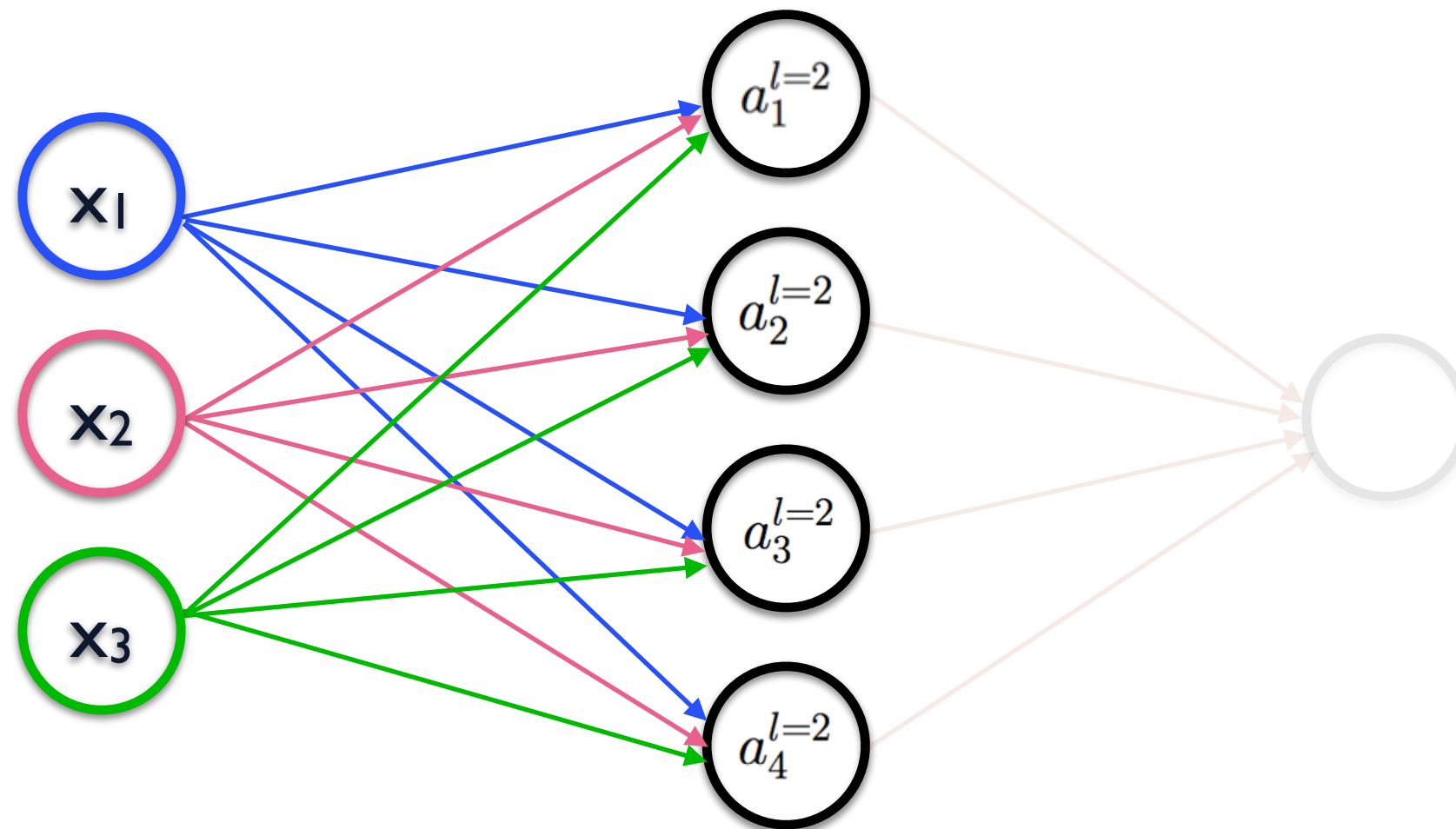
Neural Networks



Computing the output/activation of a single neuron

$$a_3^{l=2} = g\left(x_1\theta_{31}^{l=1} + x_2\theta_{32}^{l=1} + x_3\theta_{33}^{l=1}\right)$$

Neural Networks



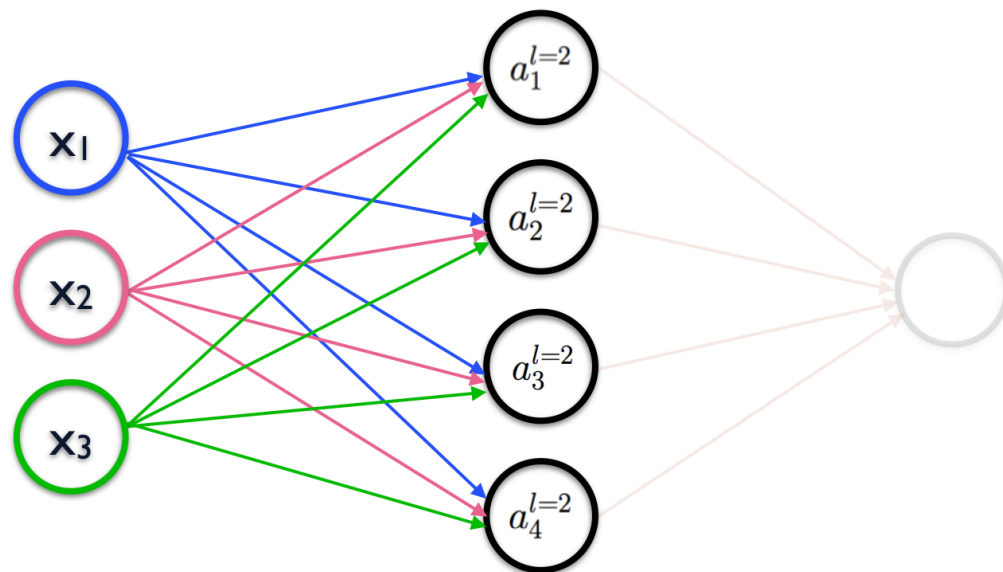
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Vectorization



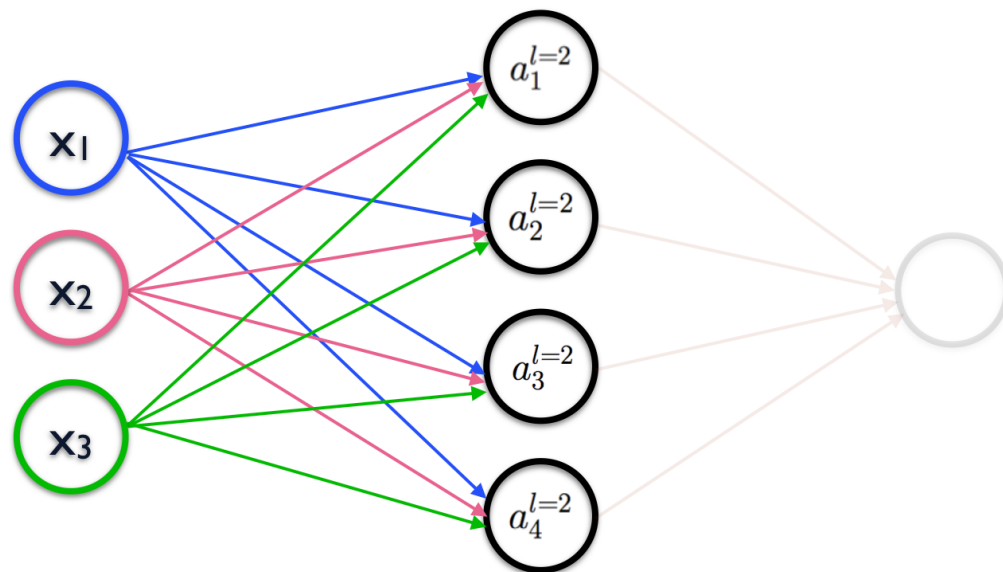
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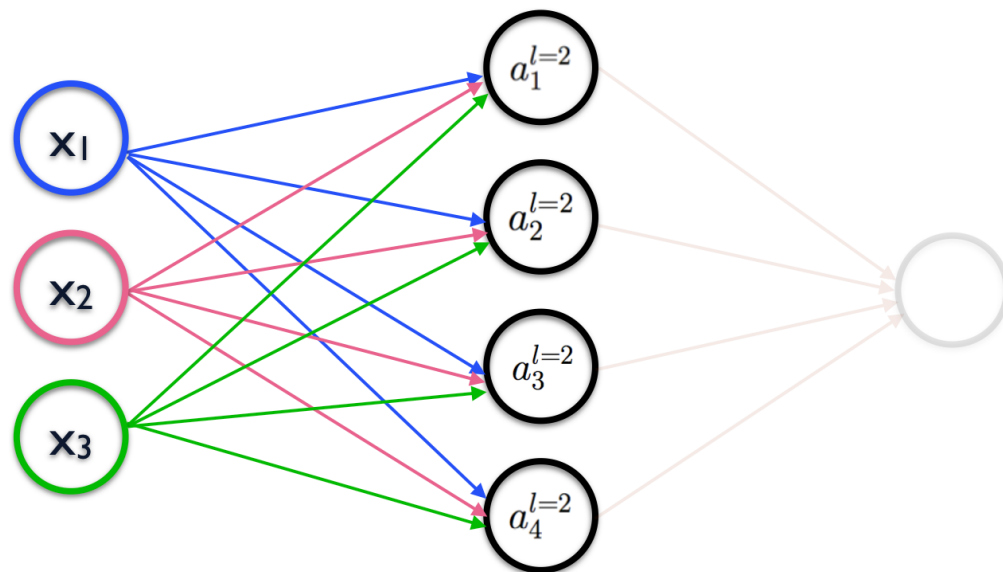
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Vectorization



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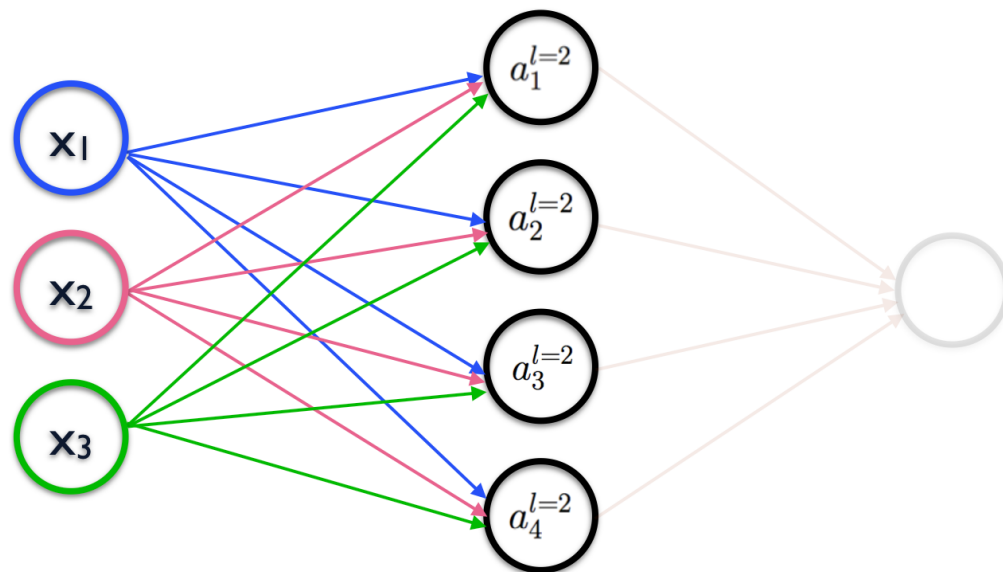
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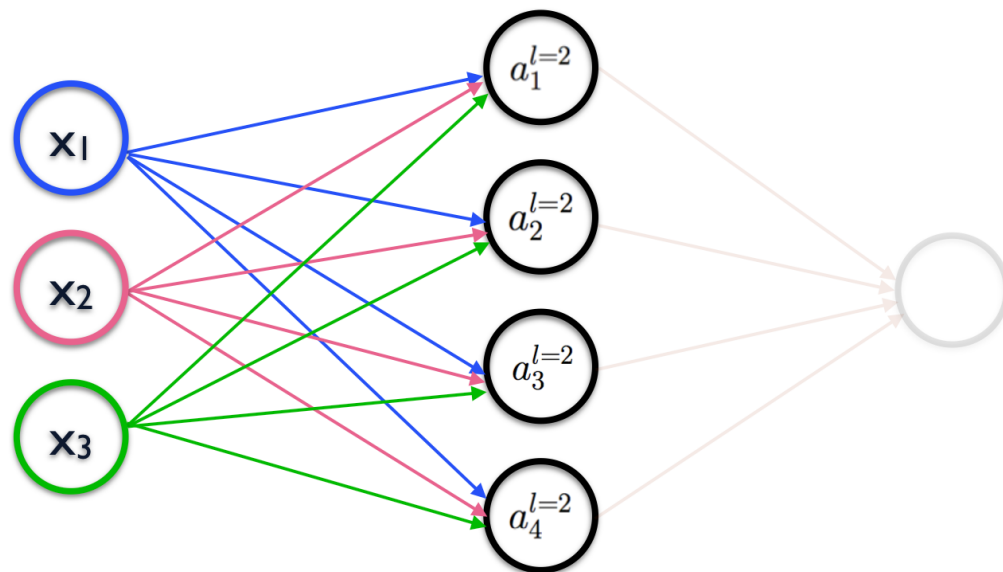
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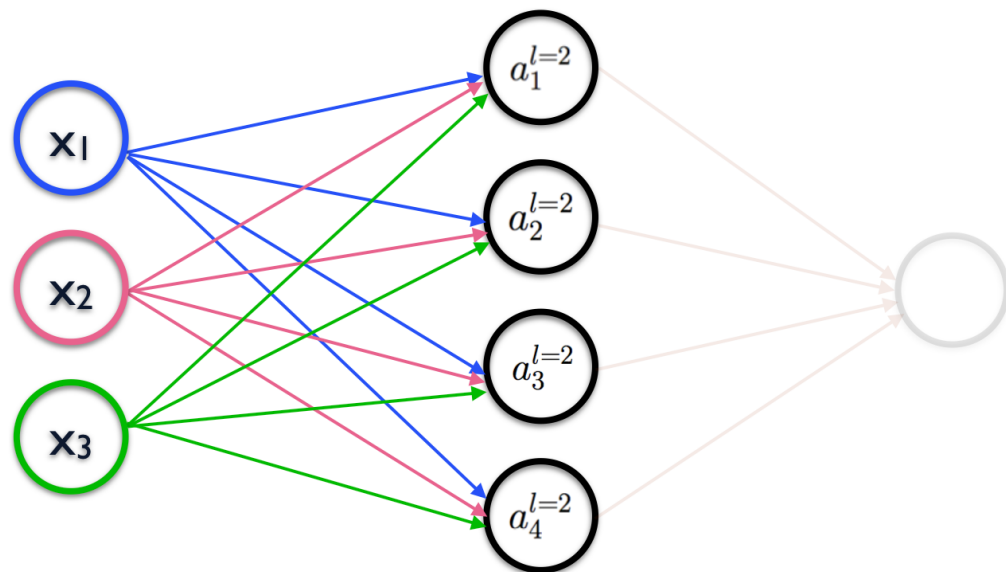
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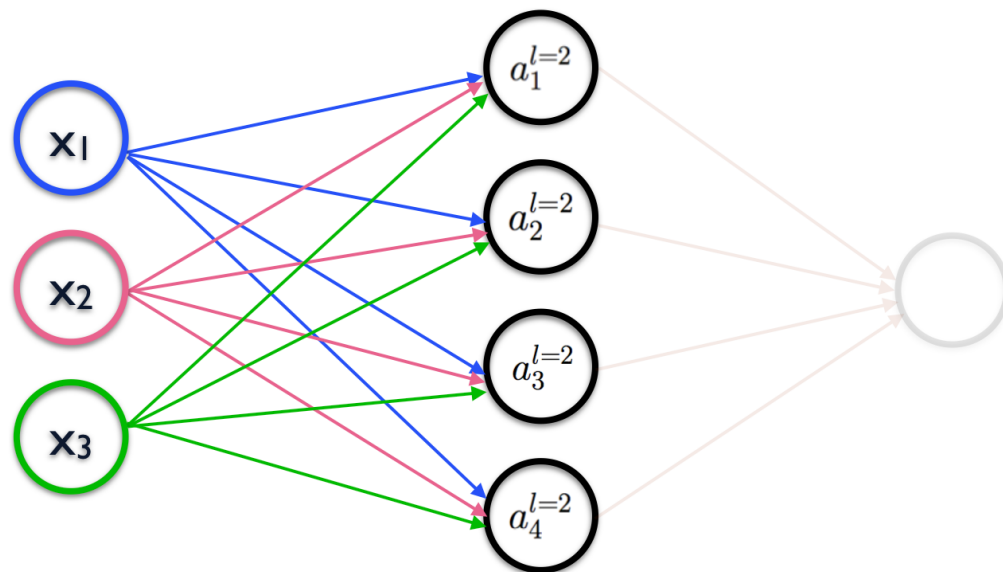
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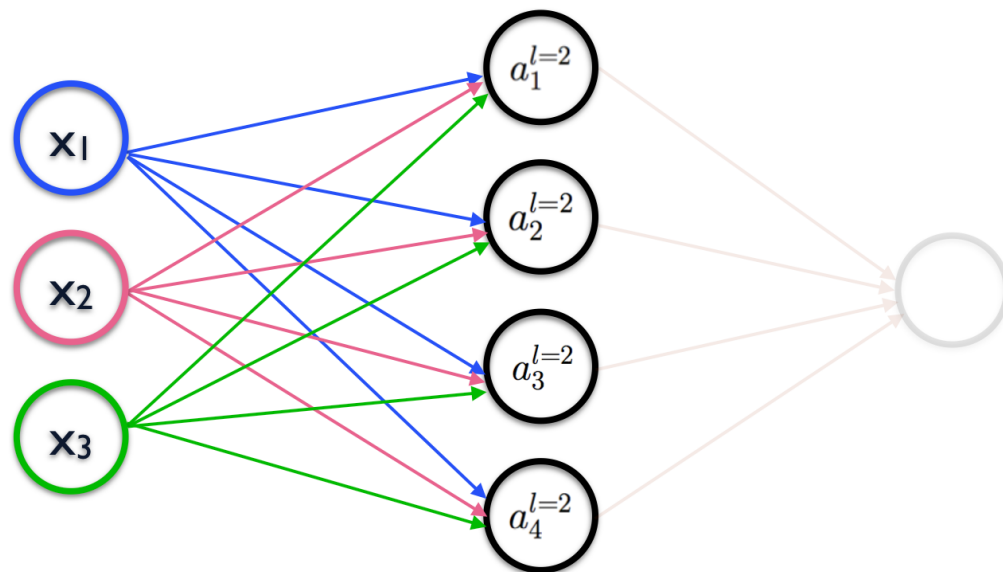
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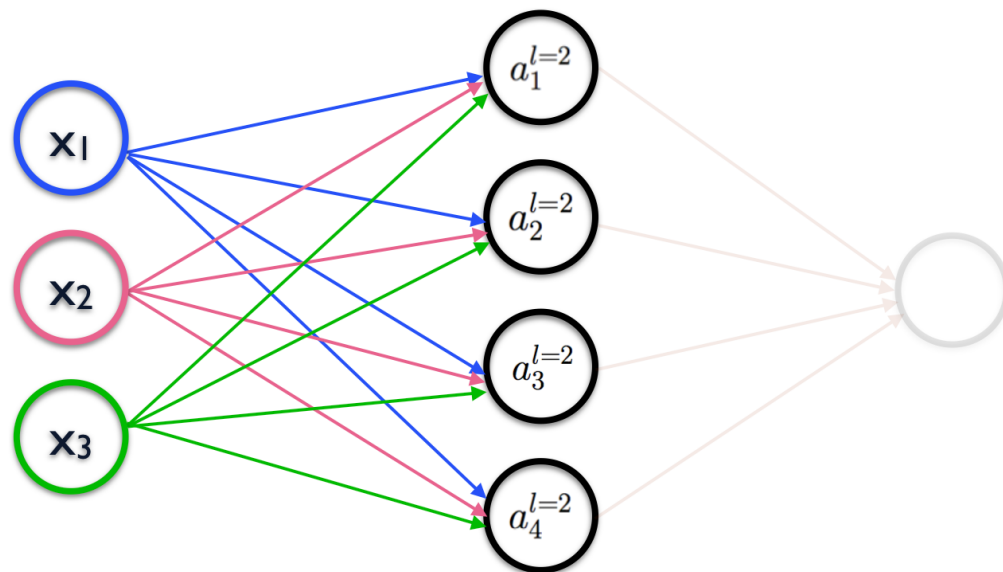
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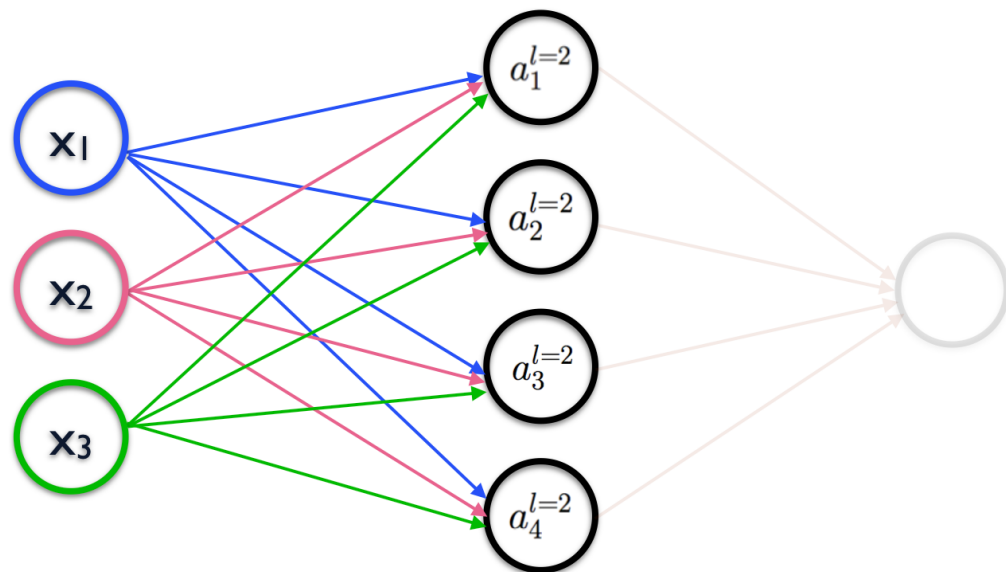
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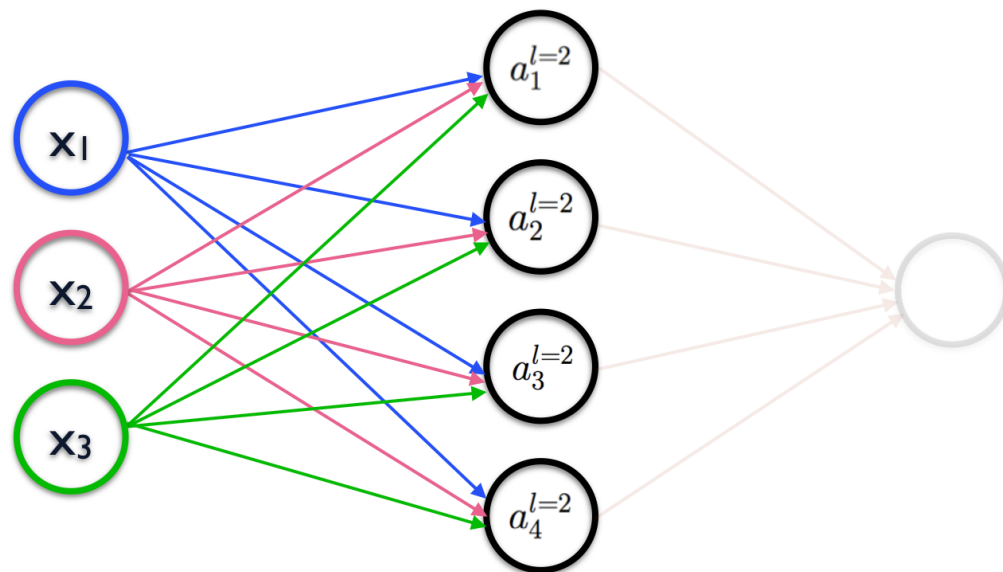
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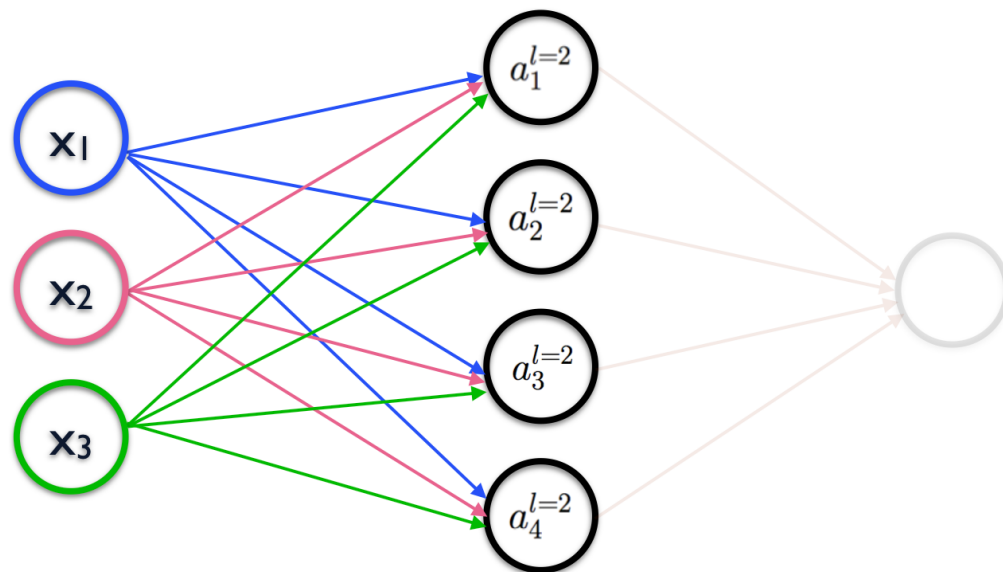
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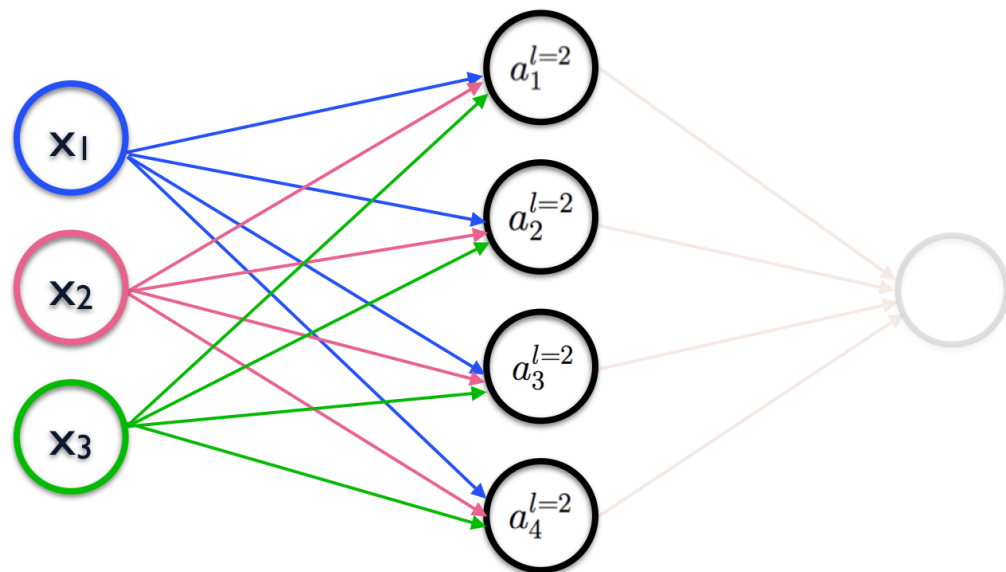
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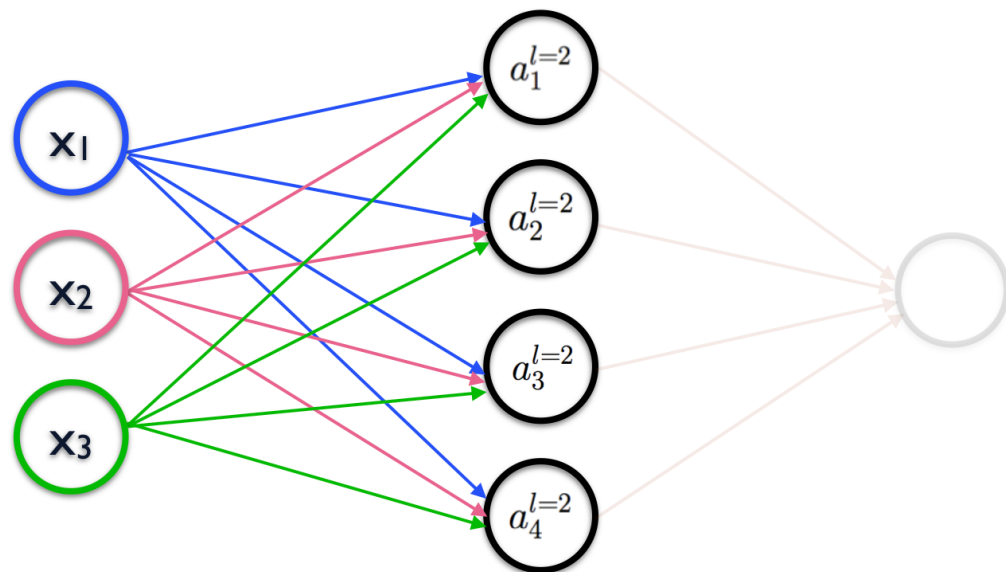
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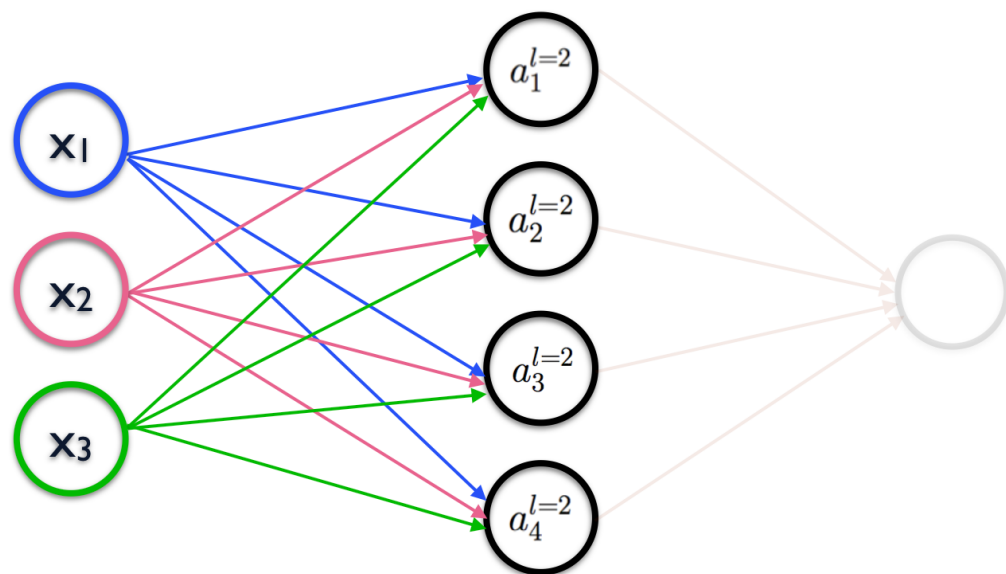
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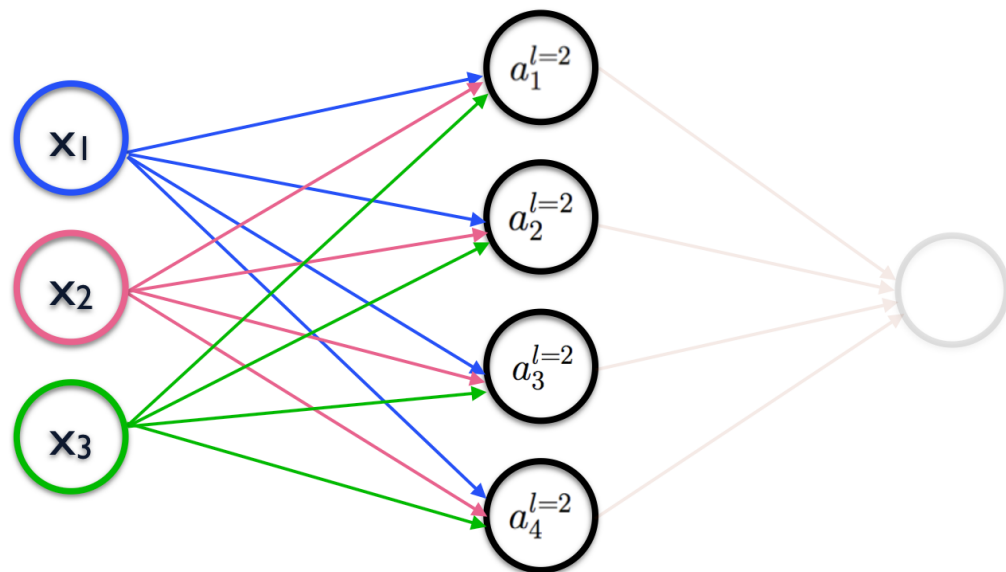
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 a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

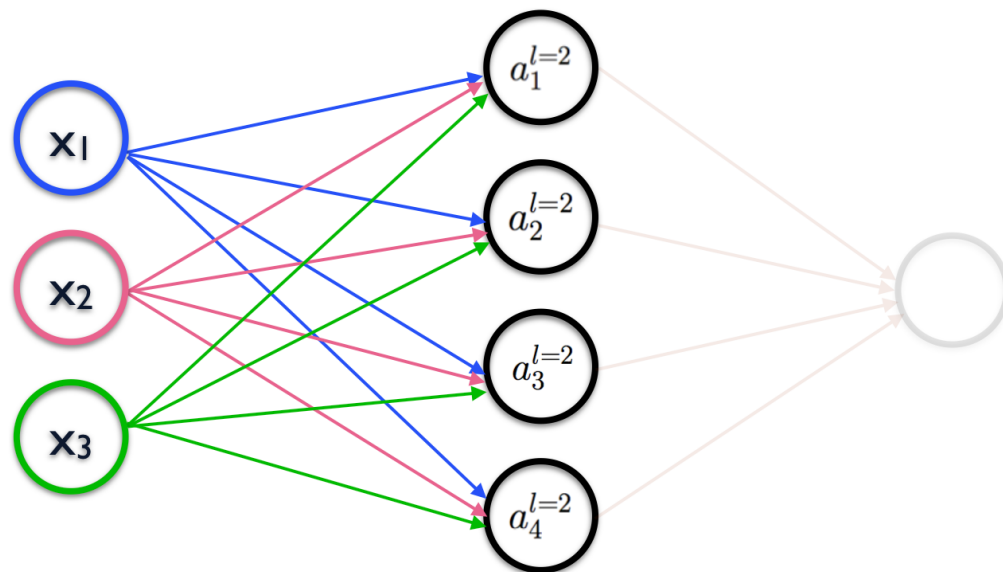
Vectorization



$$\begin{aligned}
 a_1^{l=2} &= g(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}) \\
 a_2^{l=2} &= g(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}) \\
 a_3^{l=2} &= g(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}) \\
 a_4^{l=2} &= g(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1})
 \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

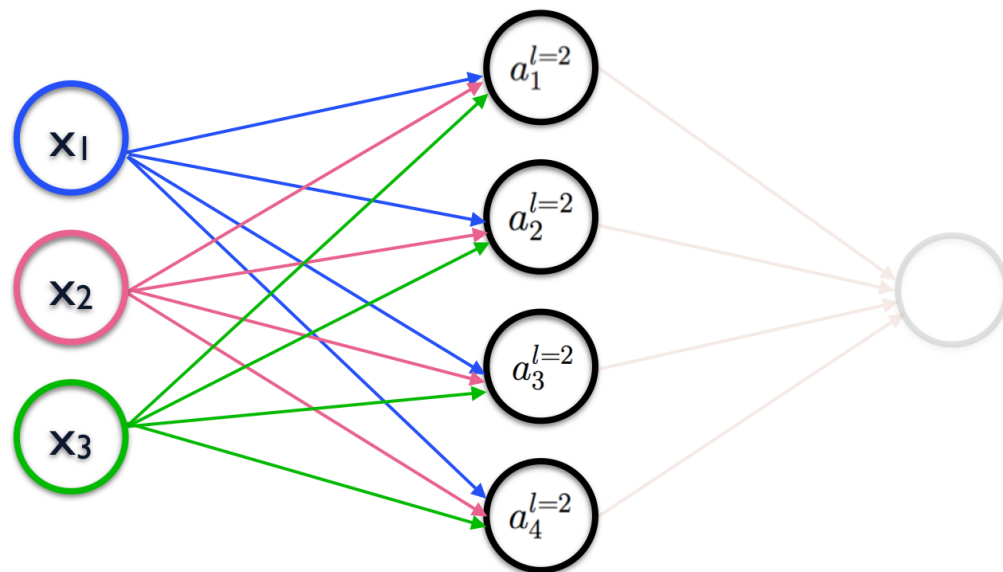
Vectorization



$$\begin{aligned}
 a_1^{l=2} &= g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

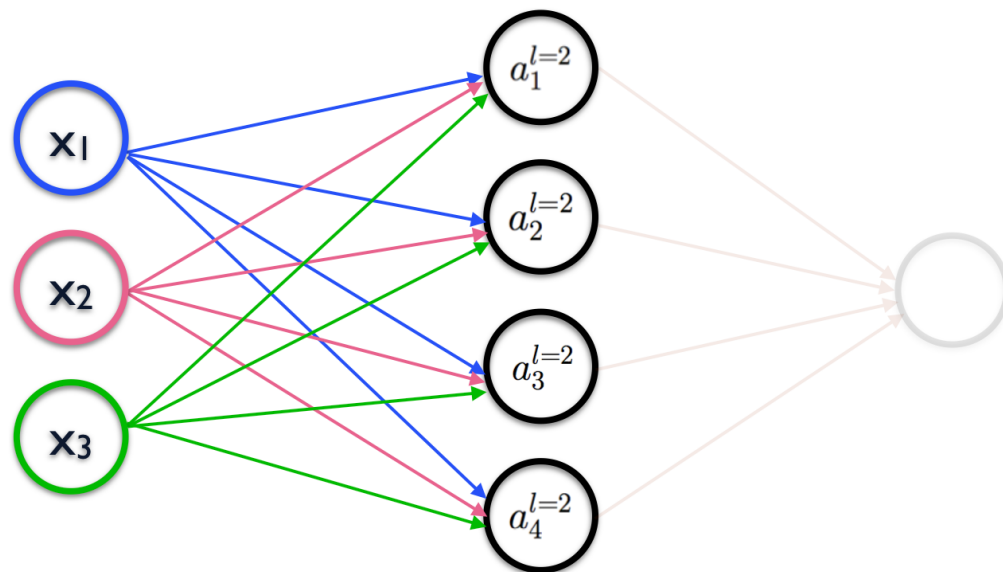
Vectorization



$$\begin{aligned}
 a_1^{l=2} &= g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

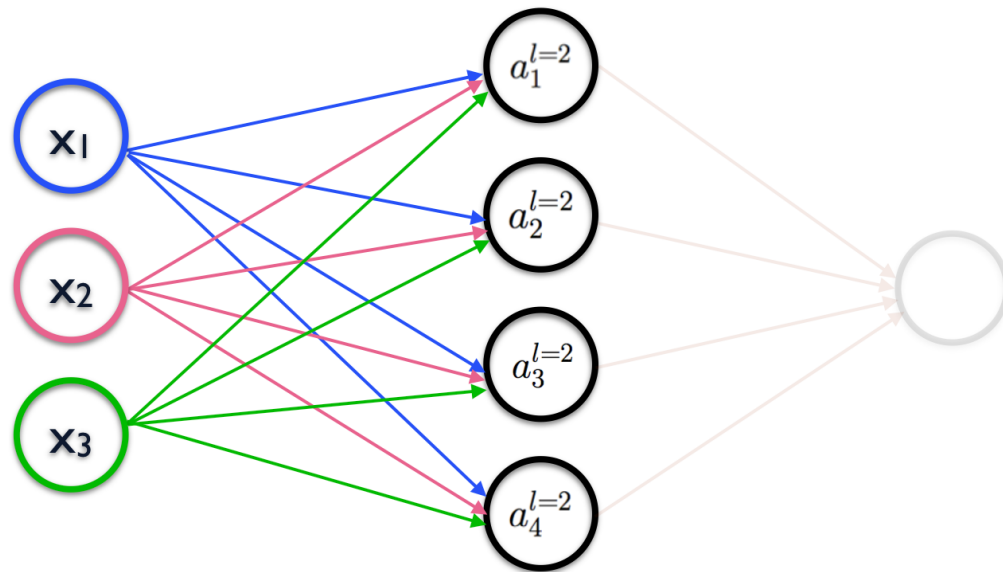
Vectorization



$$\begin{aligned}
 a_1^{l=2} &= g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

Vectorization



$$a_1^{l=2} = g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right)$$

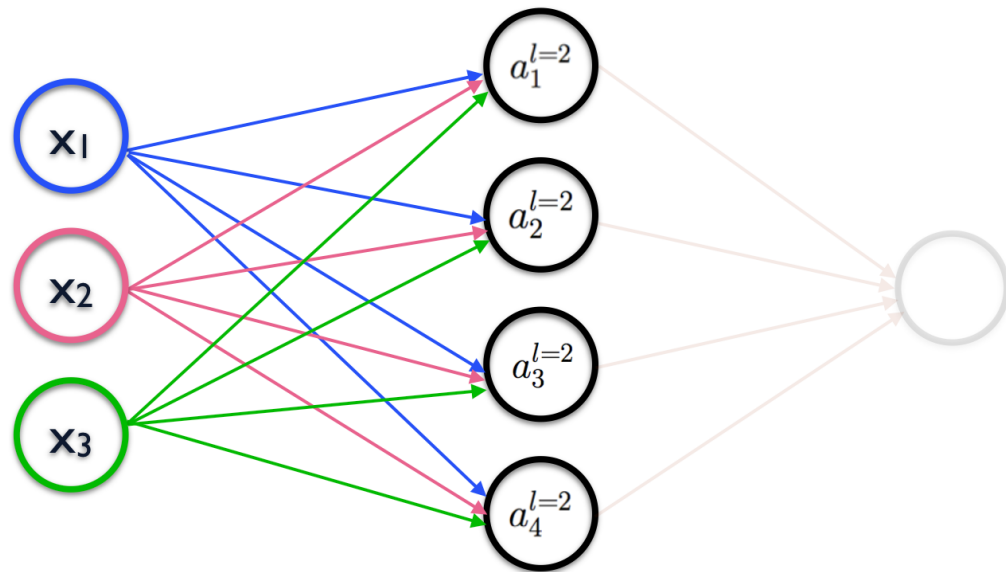
$$a_2^{l=2} = g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right)$$

$$a_3^{l=2} = g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right)$$

$$a_4^{l=2} = g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)$$

$$\begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

Vectorization



$$a_1^{l=2} = g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right)$$

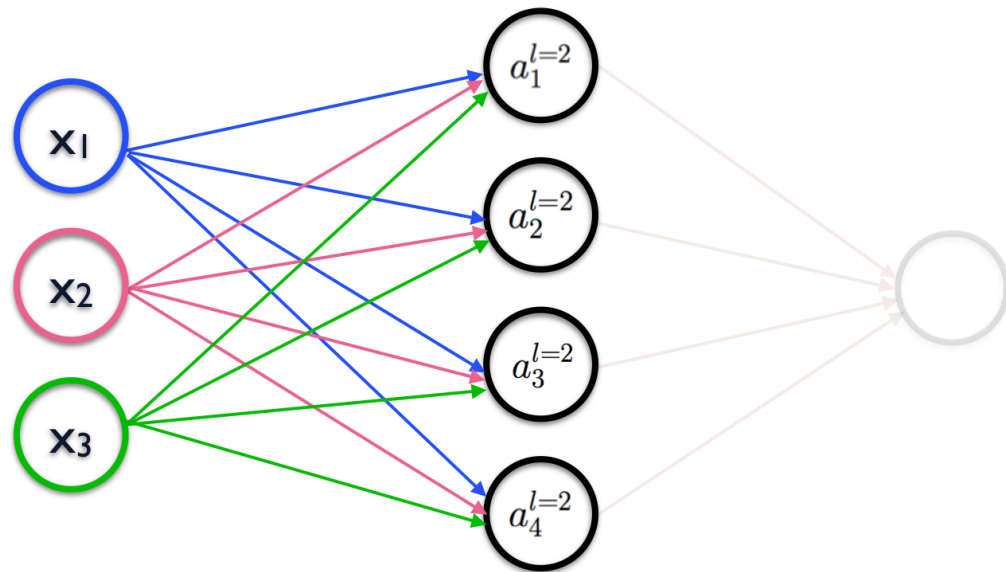
$$a_2^{l=2} = g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right)$$

$$a_3^{l=2} = g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right)$$

$$a_4^{l=2} = g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)$$

$$g\left(\begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}\right) = \begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = \begin{bmatrix} g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\ g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\ g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\ g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right) \end{bmatrix}$$

Vectorization

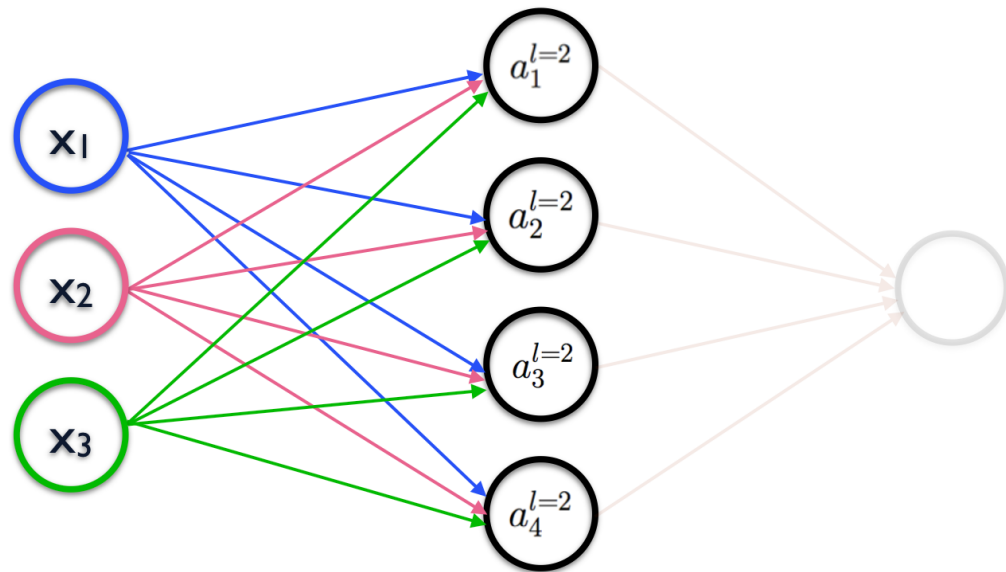


$$\begin{aligned}
 a_1^{l=2} &= g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \mathbf{x} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

$\theta^{l=1} \quad \mathbf{x} \quad x^{(i)} \quad = \quad z^{l=2}$

Vectorization



$$\begin{aligned}
 a_1^{l=2} &= g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \mathbf{x} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix}$$

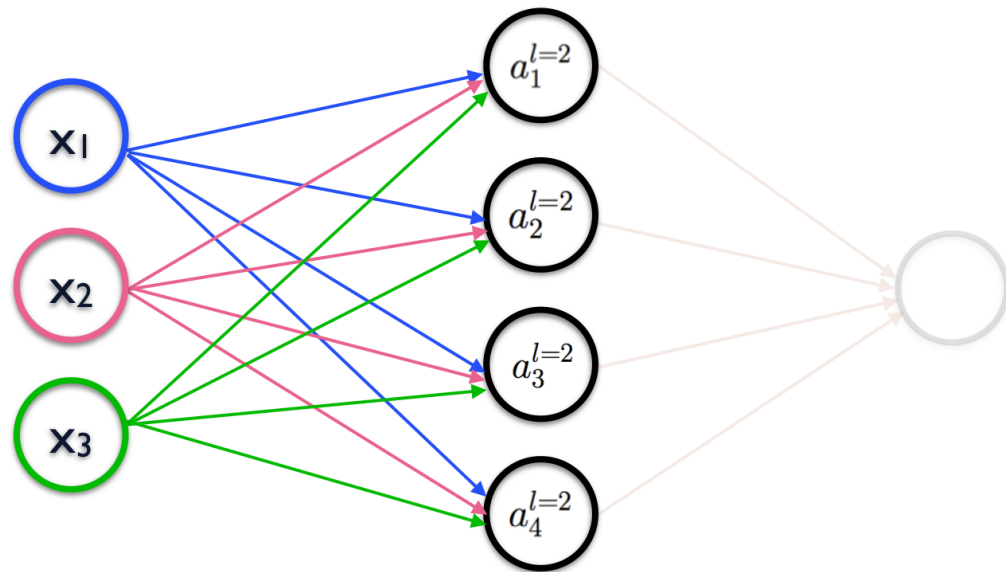
$$\theta^{l=1} \in \mathbb{R}^{4 \times 3}$$

$$x^{(i)} \in \mathbb{R}^{3 \times 1}$$

$$z^{l=2} \in \mathbb{R}^{4 \times 1}$$



Vectorization



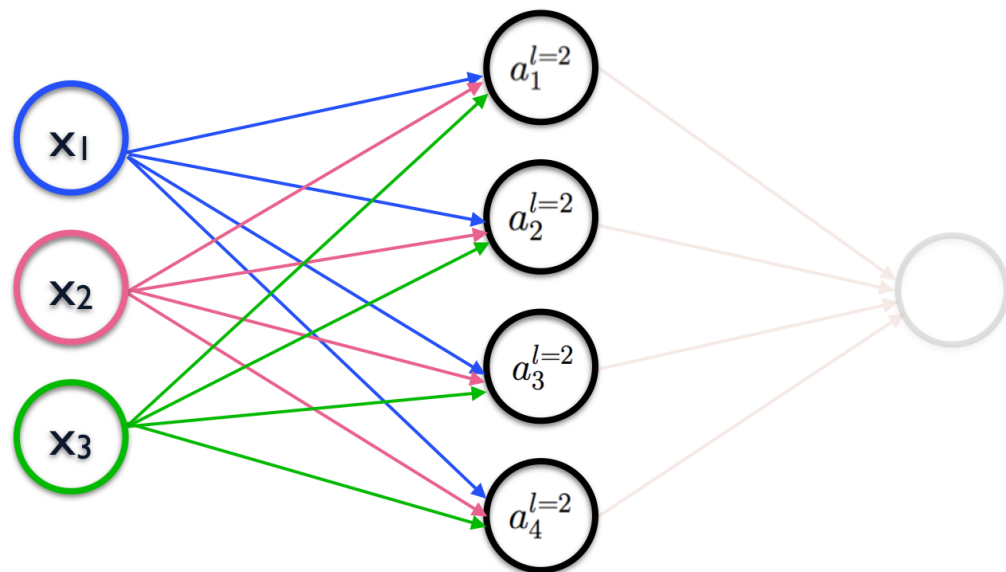
$$\begin{aligned} a_1^{l=2} &= g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\ a_2^{l=2} &= g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\ a_3^{l=2} &= g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\ a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right) \end{aligned}$$

$$\begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = g \left(\begin{bmatrix} x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{bmatrix} \right)$$

$z^{l=2} \in \mathbb{R}^{4 \times 1}$

$a^{l=2} \in \mathbb{R}^{4 \times 1}$

Vectorization

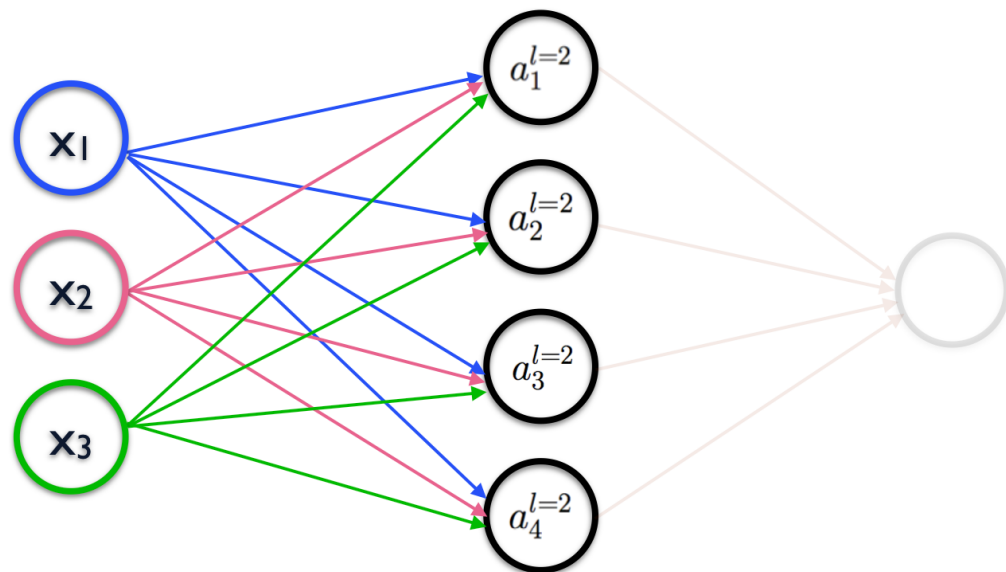


$$\begin{aligned}
 a_1^{l=2} &= g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = g \left(\begin{matrix} z^{l=2} \\ \theta^{l=1} \mathbf{x} x^{(i)} \end{matrix} \right)$$

$a^{l=2} \in \mathbb{R}^{4 \times 1}$

Vectorization



$$a_1^{l=2} = g\left(x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right)$$

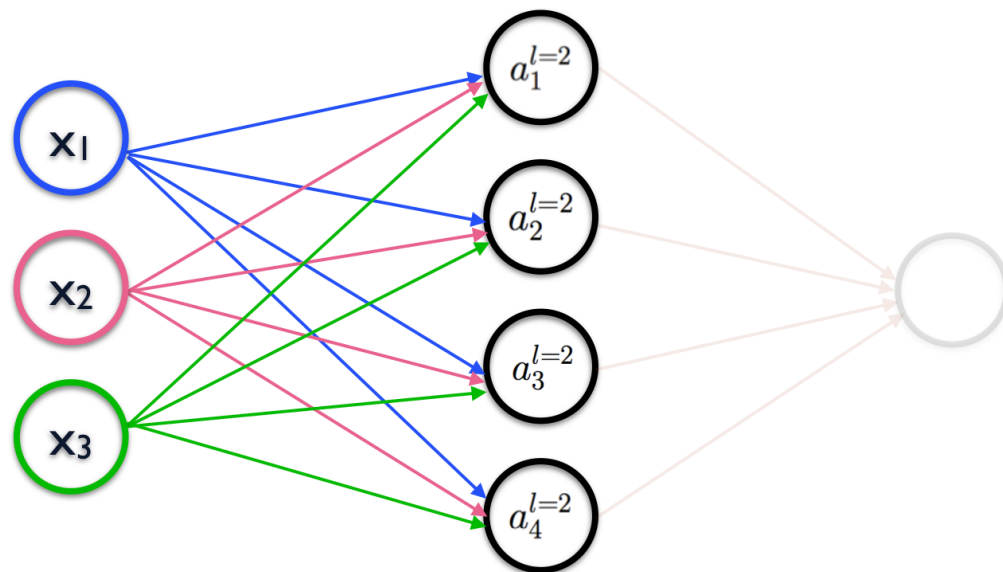
$$a_2^{l=2} = g\left(x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right)$$

$$a_3^{l=2} = g\left(x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right)$$

$$a_4^{l=2} = g\left(x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)$$

$$a^{l=2} = g\left(\begin{matrix} z^{l=2} \\ \theta^{l=1} \mathbf{x} \end{matrix} x^{(i)} \right)$$

Vectorization



$$a_1^{l=2} = g\left(x_1\theta_{11}^{l=1} + x_2\theta_{12}^{l=1} + x_3\theta_{13}^{l=1}\right)$$

$$a_2^{l=2} = g\left(x_1\theta_{21}^{l=1} + x_2\theta_{22}^{l=1} + x_3\theta_{23}^{l=1}\right)$$

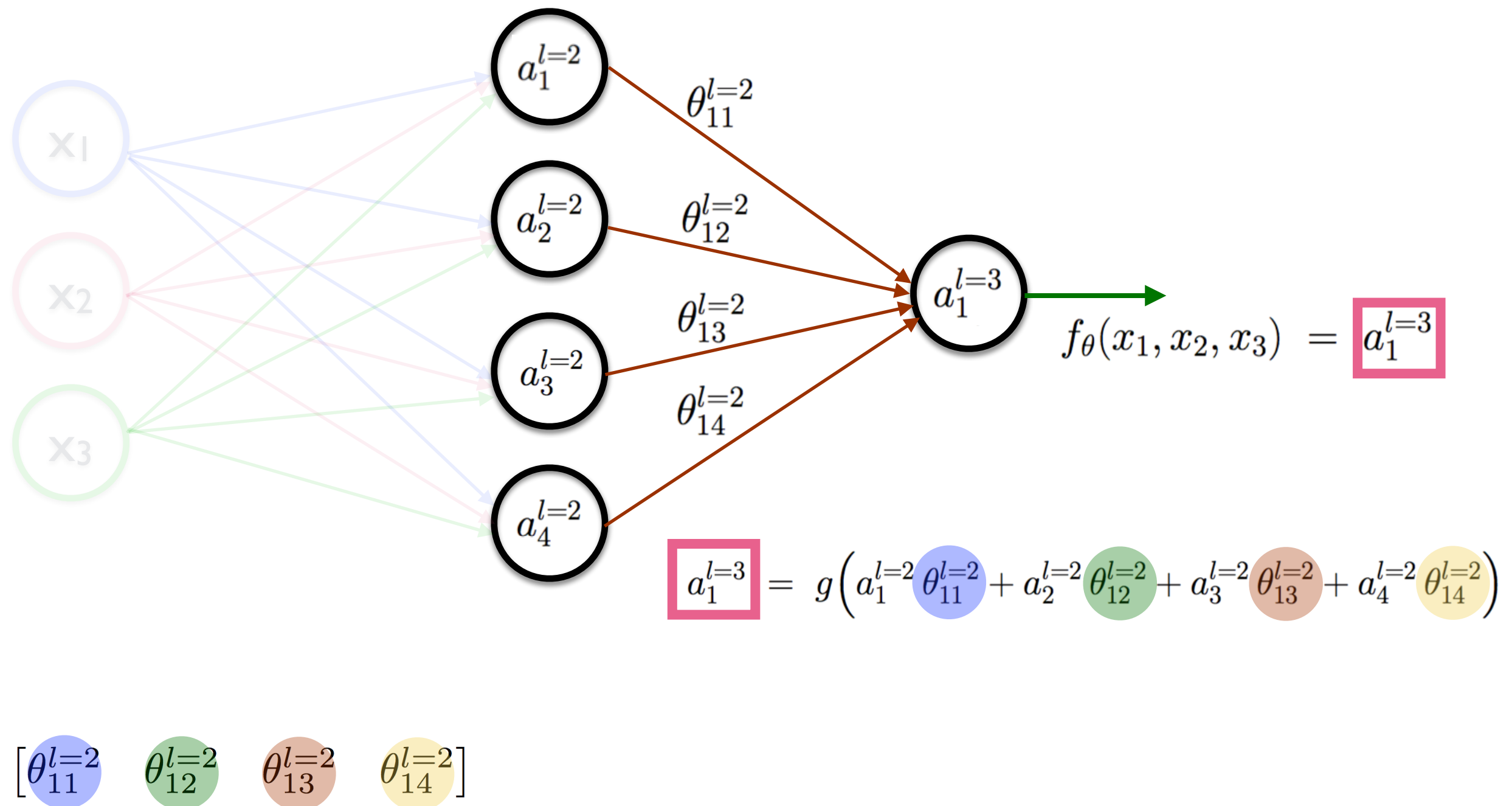
$$a_3^{l=2} = g\left(x_1\theta_{31}^{l=1} + x_2\theta_{32}^{l=1} + x_3\theta_{33}^{l=1}\right)$$

$$a_4^{l=2} = g\left(x_1\theta_{41}^{l=1} + x_2\theta_{42}^{l=1} + x_3\theta_{43}^{l=1}\right)$$

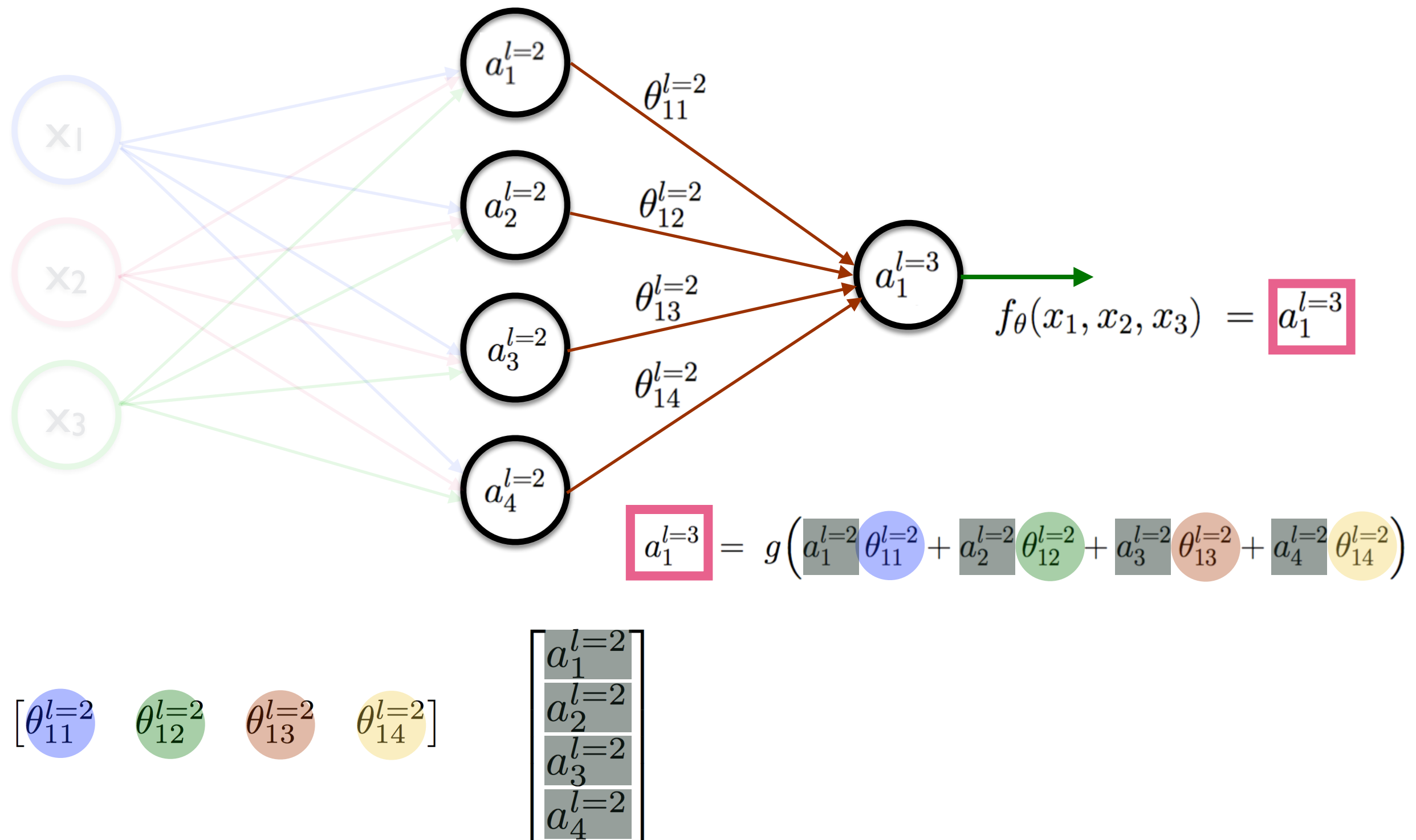
$$z^{l=2} = \theta^{l=1} \mathbf{x} x^{(i)}$$

$$a^{l=2} = g\left(z^{l=2}\right)$$

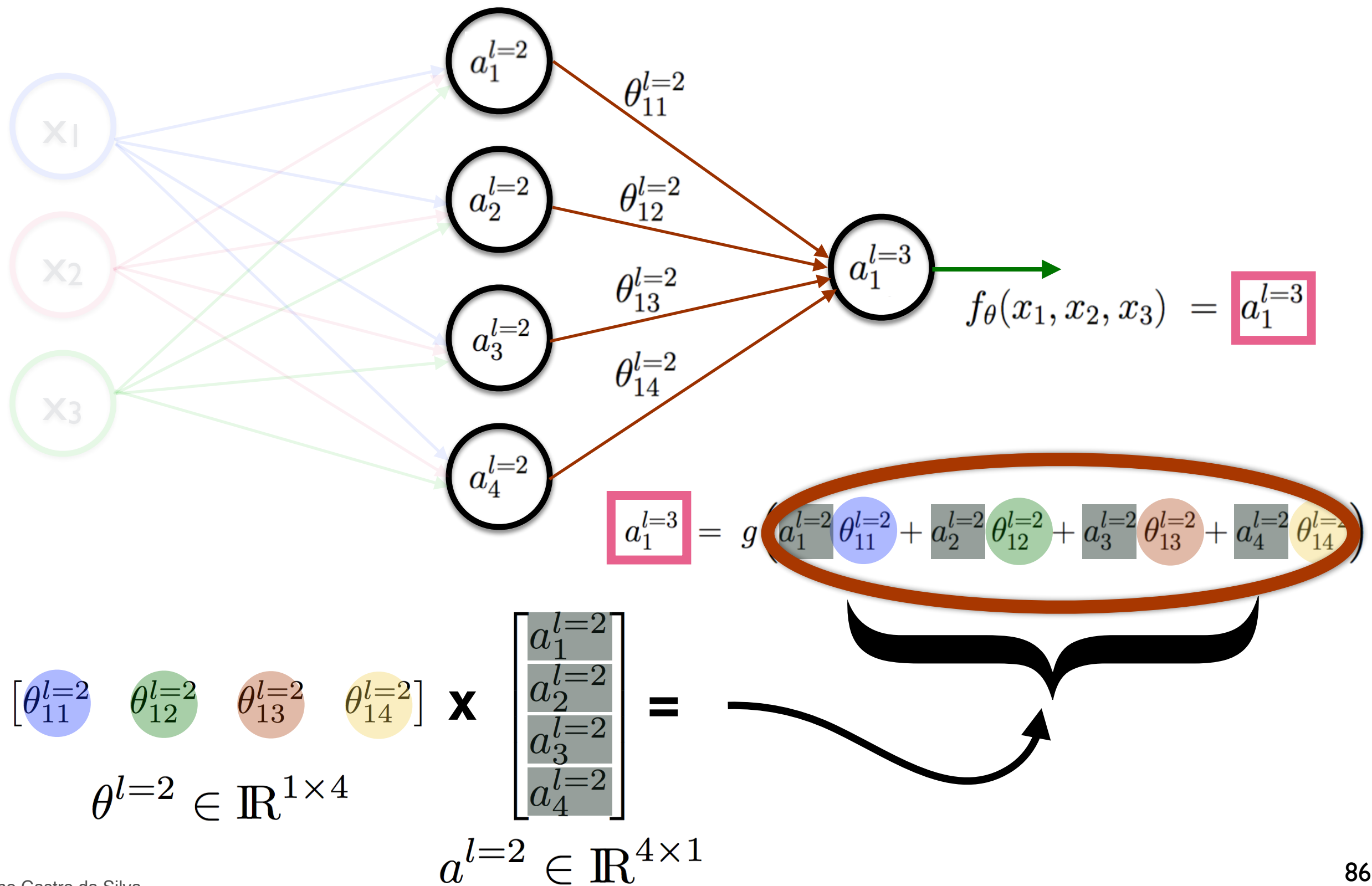
Vectorization



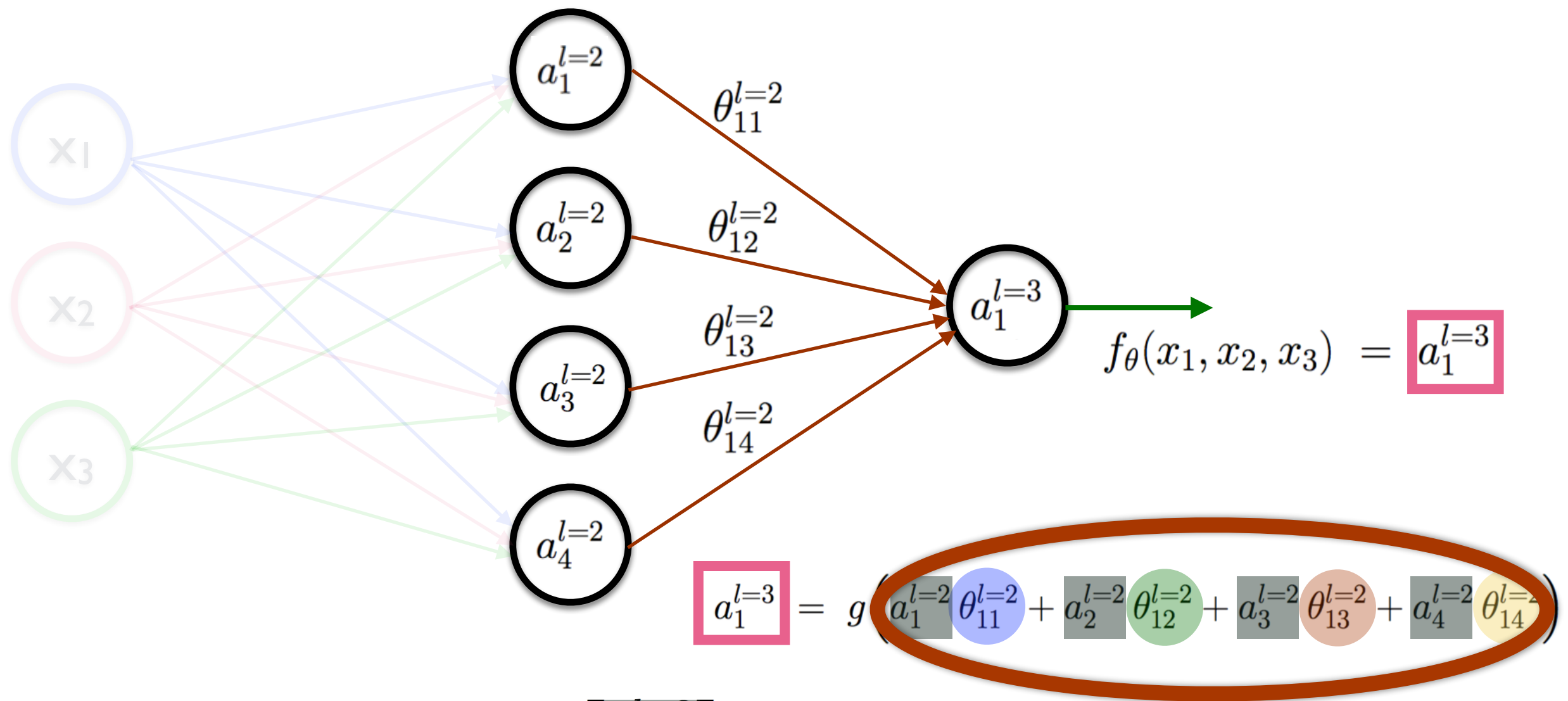
Vectorization



Vectorization



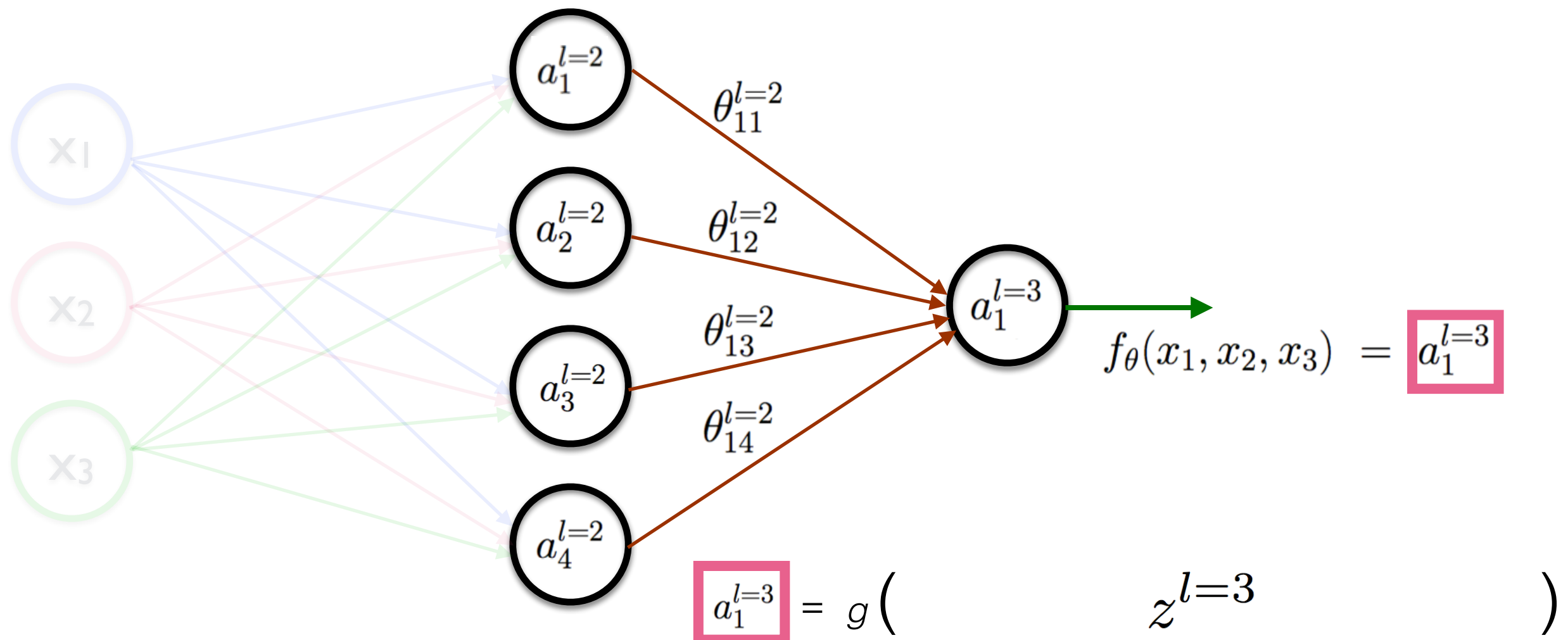
Vectorization



$$[\theta_{11}^{l=2} \quad \theta_{12}^{l=2} \quad \theta_{13}^{l=2} \quad \theta_{14}^{l=2}] \mathbf{x} \begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

$\theta^{l=2} \in \mathbb{R}^{1 \times 4}$ $a^{l=2} \in \mathbb{R}^{4 \times 1}$

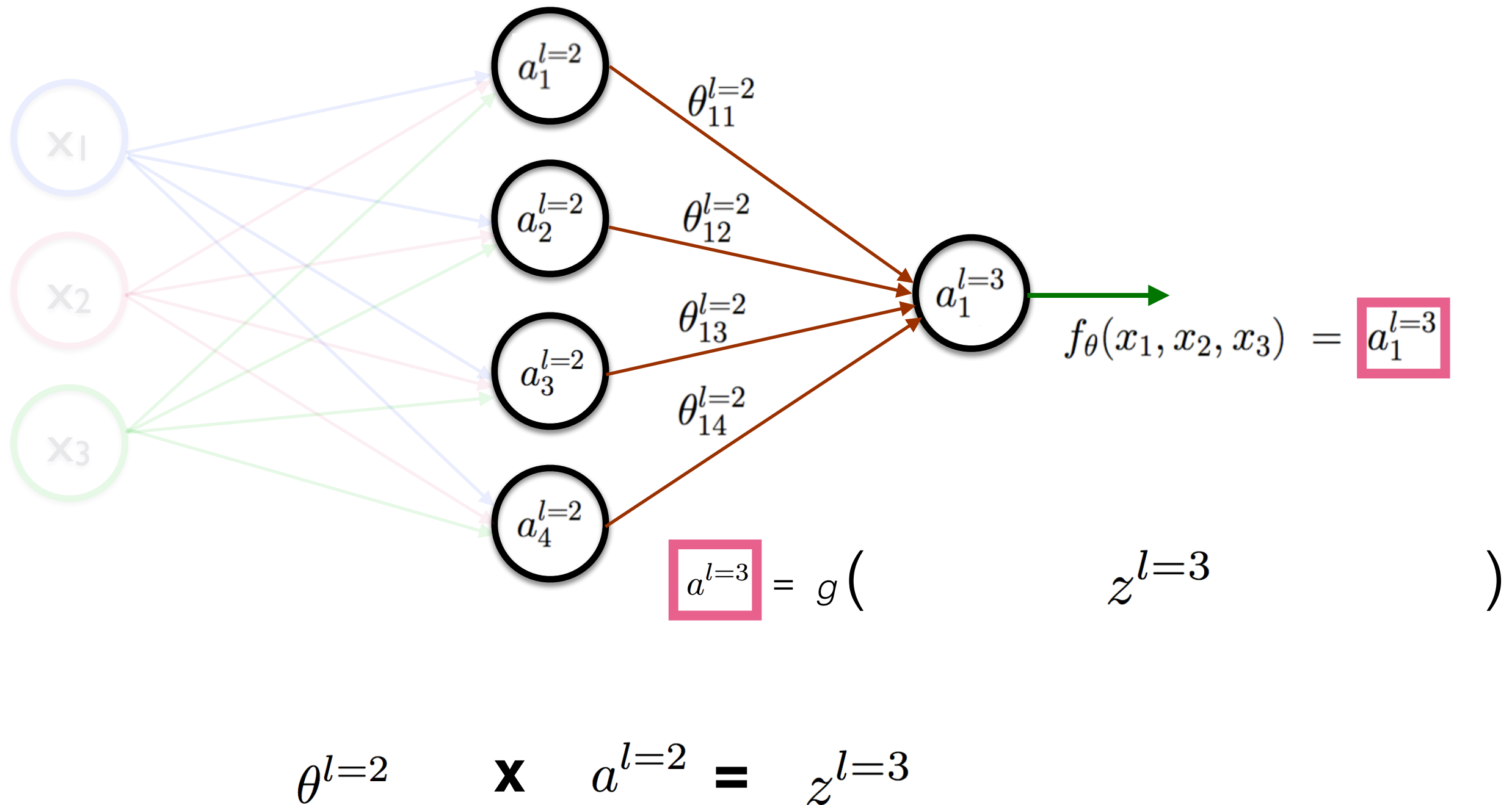
Vectorization



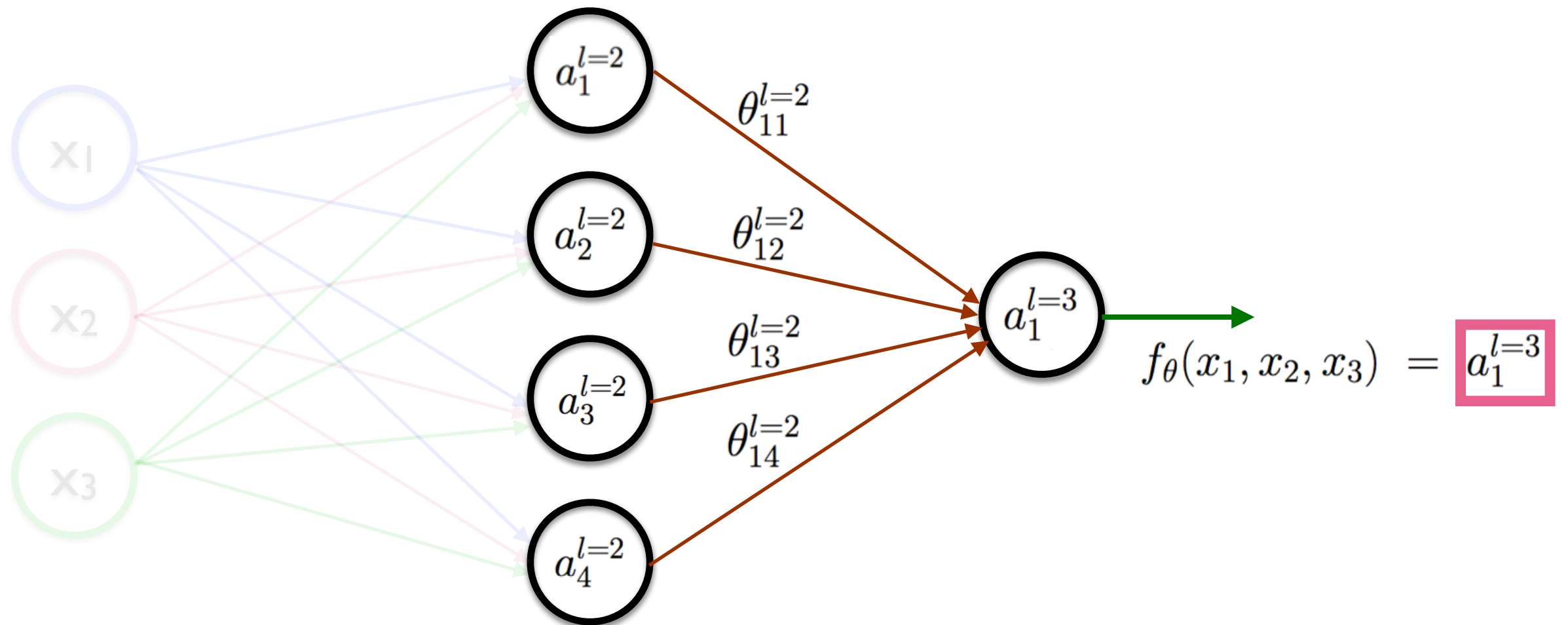
$$\begin{bmatrix} \theta_{11}^{l=2} & \theta_{12}^{l=2} & \theta_{13}^{l=2} & \theta_{14}^{l=2} \end{bmatrix} \mathbf{x} \begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

$\theta^{l=2} \in \mathbb{R}^{1 \times 4}$
 $a^{l=2} \in \mathbb{R}^{4 \times 1}$

Vectorization



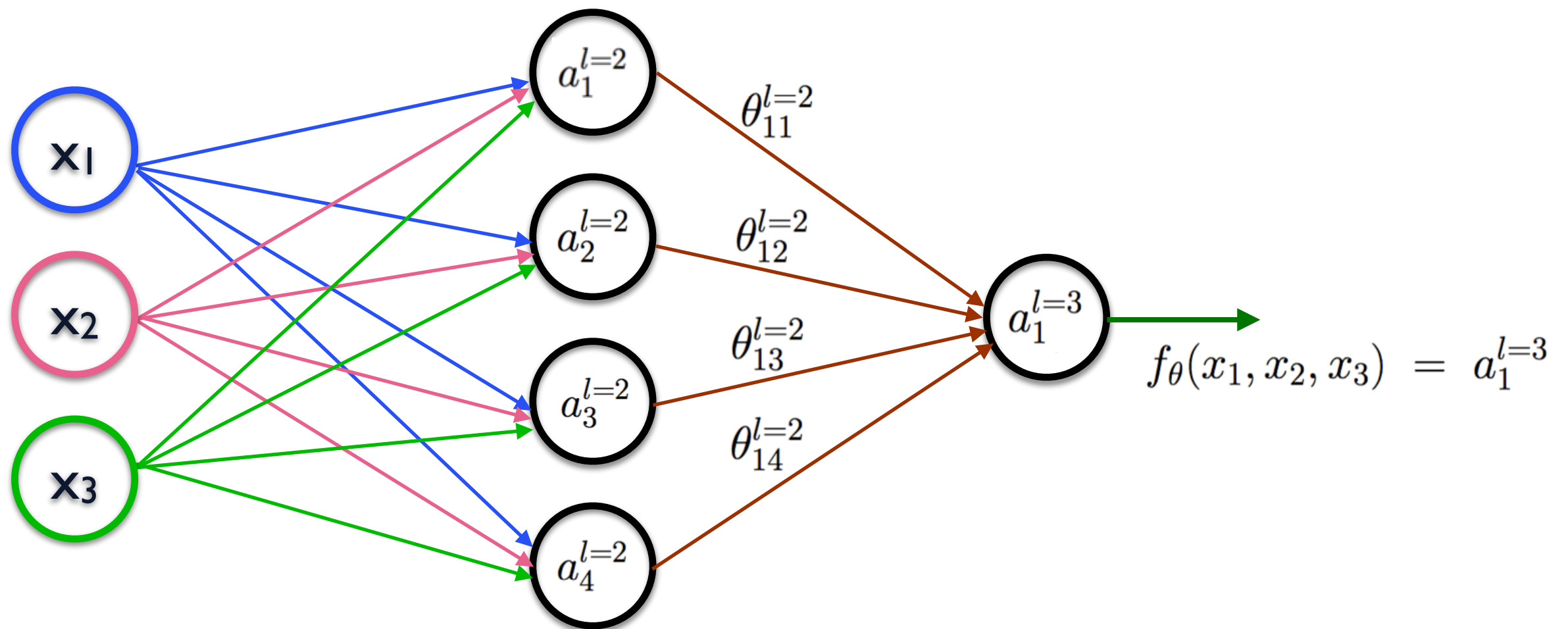
Vectorization



$$z^{l=3} = \theta^{l=2} \mathbf{x} a^{l=2}$$

$$a^{l=3} = g \left(z^{l=3} \right)$$

Vectorization



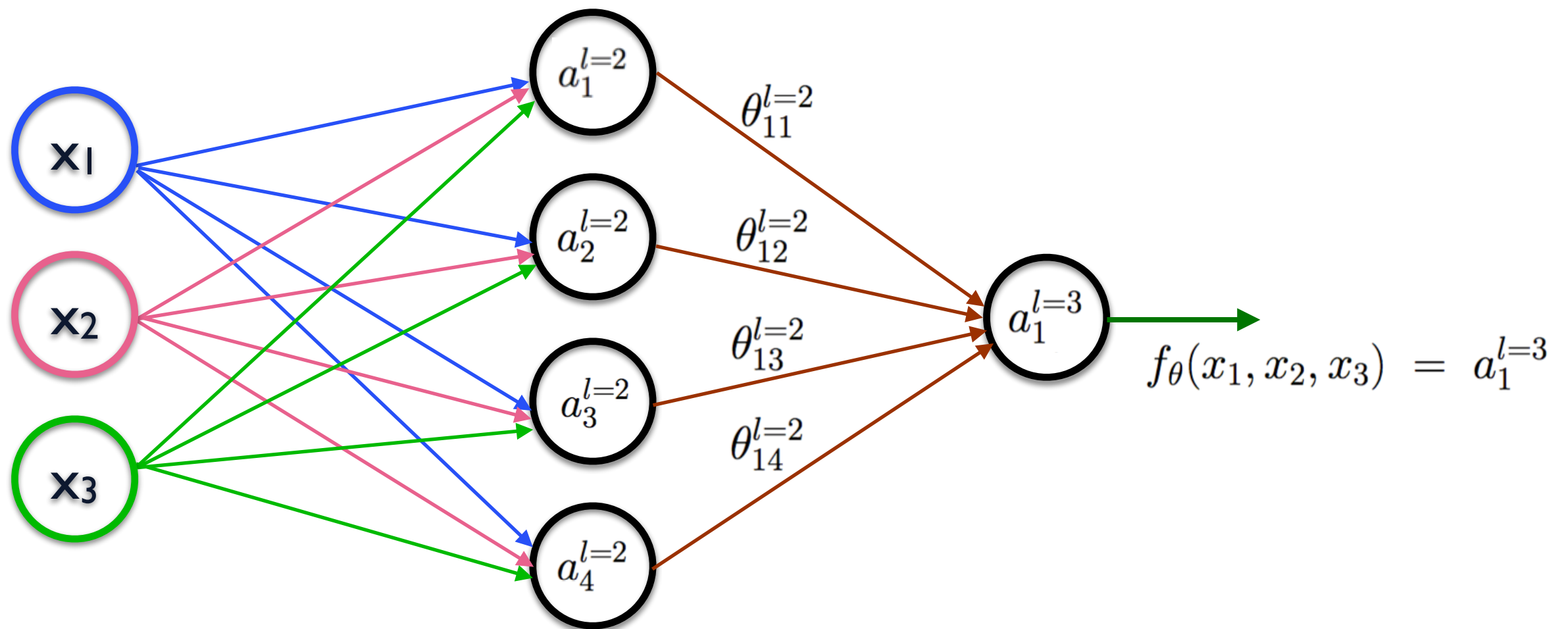
$$z^{l=2} = \theta^{l=1} \mathbf{x} x^{(i)}$$

$$z^{l=3} = \theta^{l=2} \mathbf{x} a^{l=2}$$

$$a^{l=2} = g \left(z^{l=2} \right)$$

$$a^{l=3} = g \left(z^{l=3} \right)$$

Vectorization



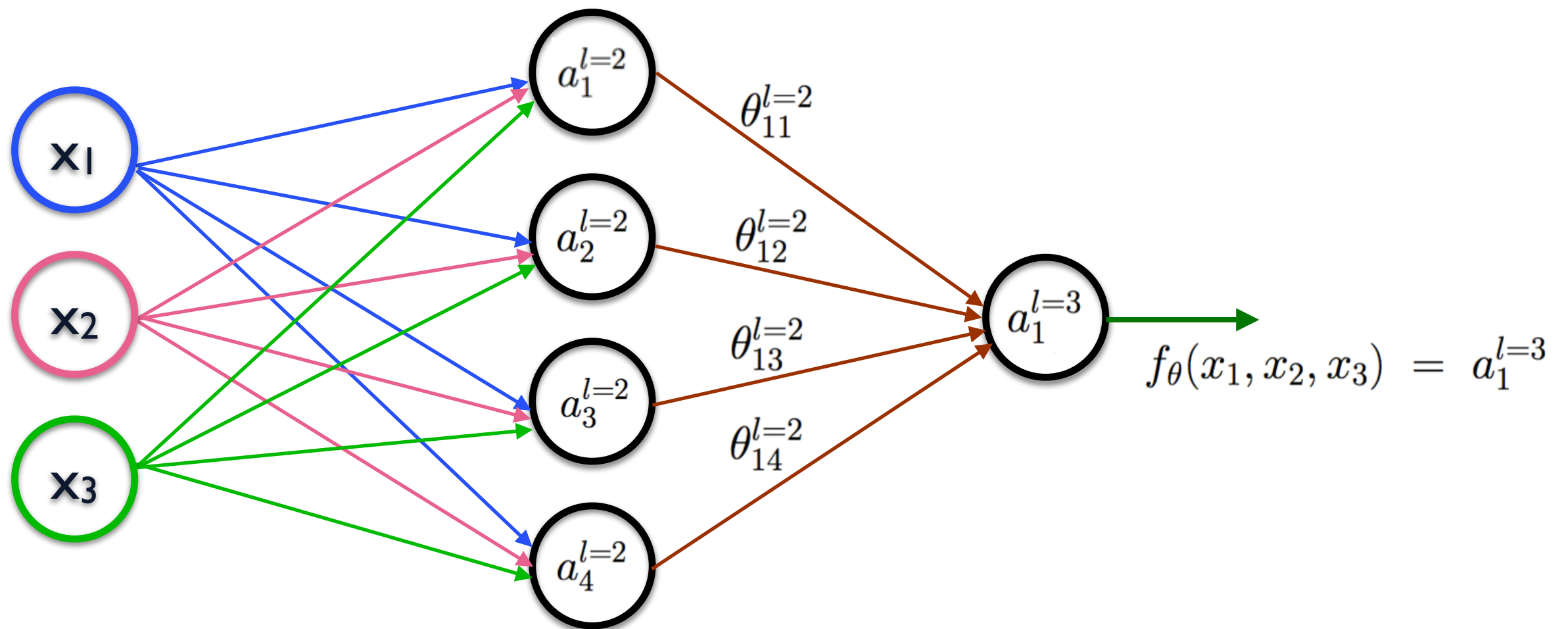
$$z^{l=2} = \theta^{l=1} \mathbf{x}^{(i)} = a^{l=1}$$

$$z^{l=3} = \theta^{l=2} \mathbf{x}^{(i)} = a^{l=2}$$

$$a^{l=2} = g(z^{l=2})$$

$$a^{l=3} = g(z^{l=3})$$

Vectorization



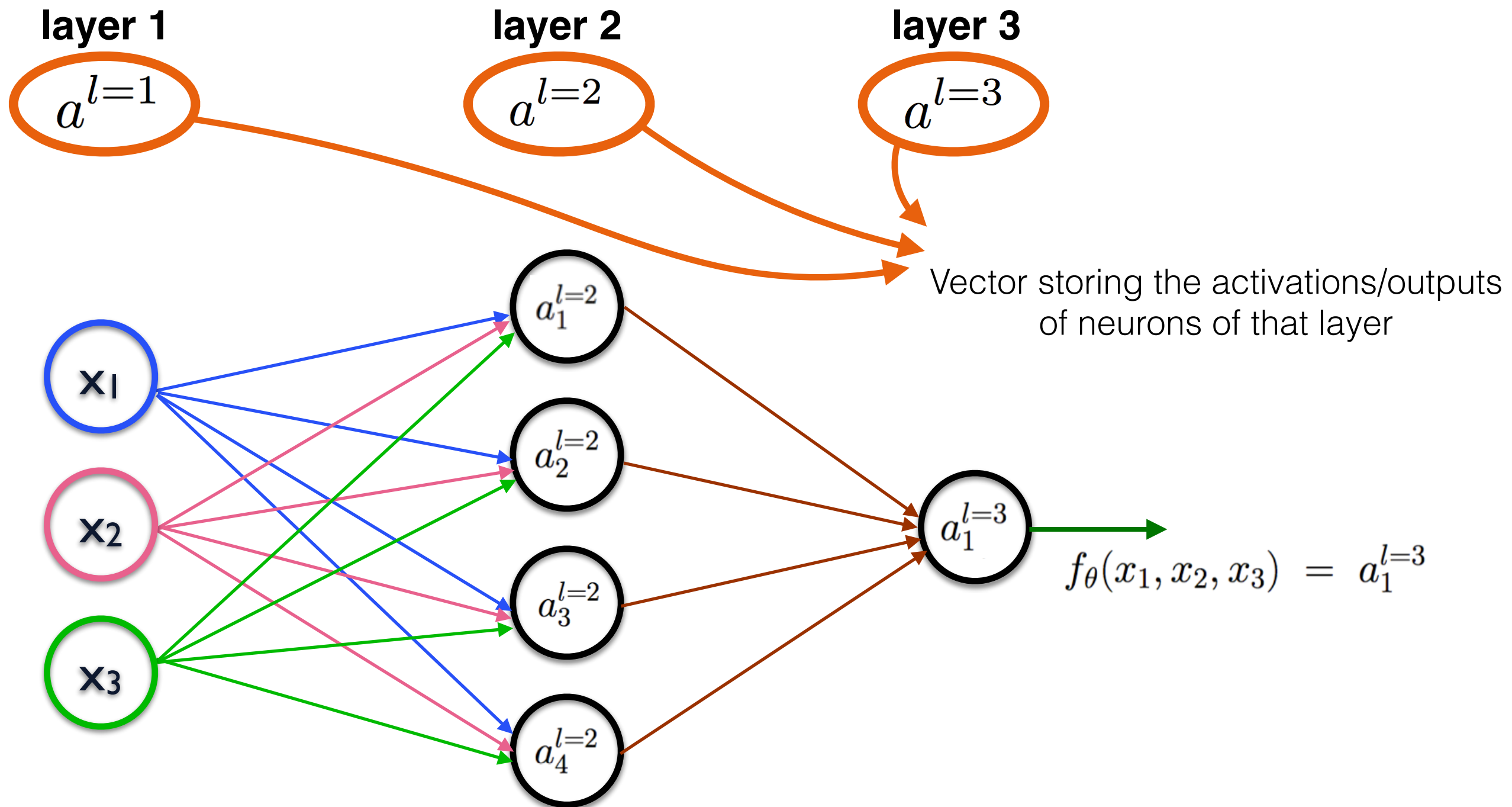
$$z^{l=2} = \theta^{l=1} \mathbf{x} a^{l=1}$$

$$z^{l=3} = \theta^{l=2} \mathbf{x} a^{l=2}$$

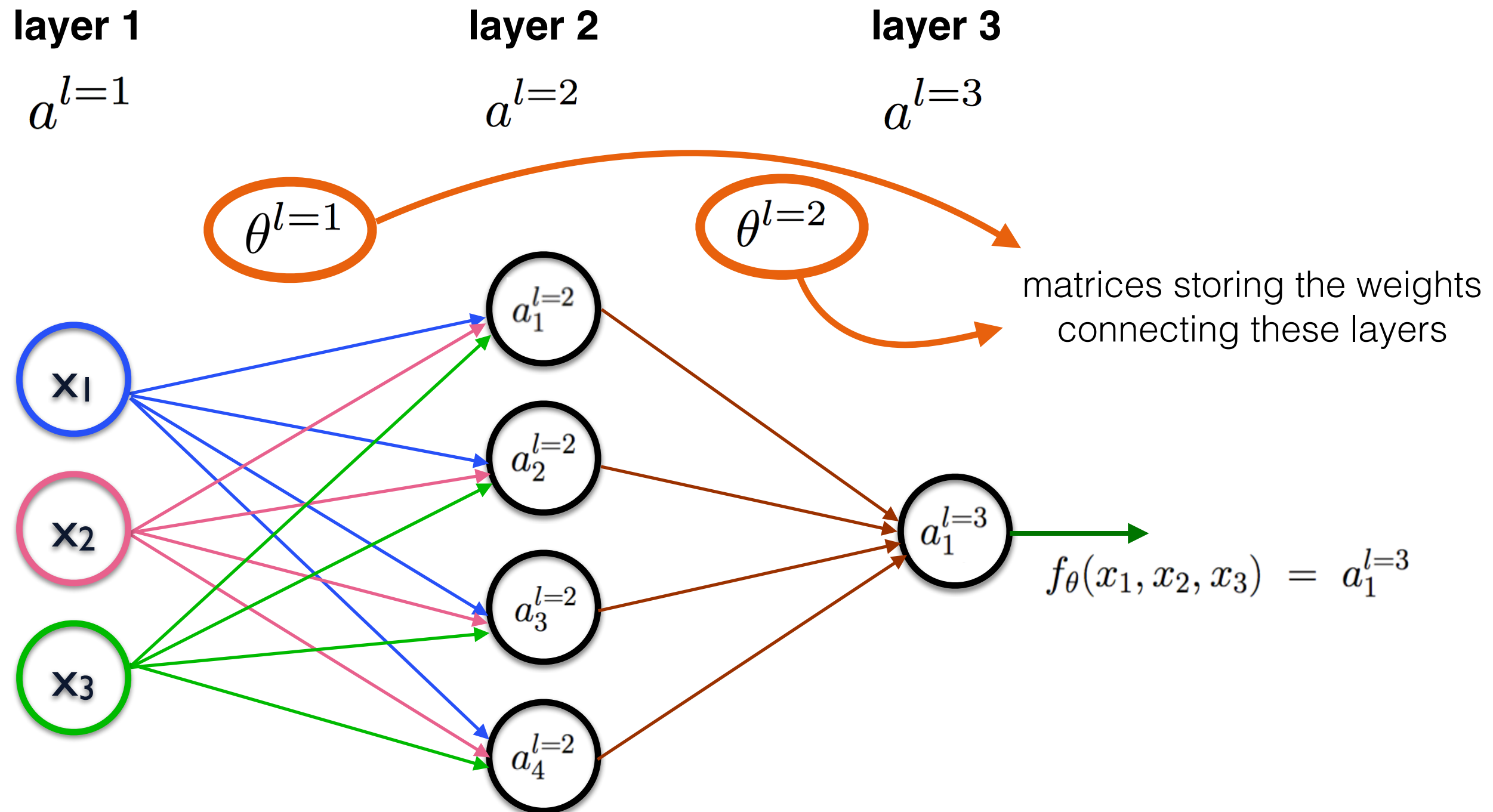
$$a^{l=2} = g \left(z^{l=2} \right)$$

$$a^{l=3} = g \left(z^{l=3} \right)$$

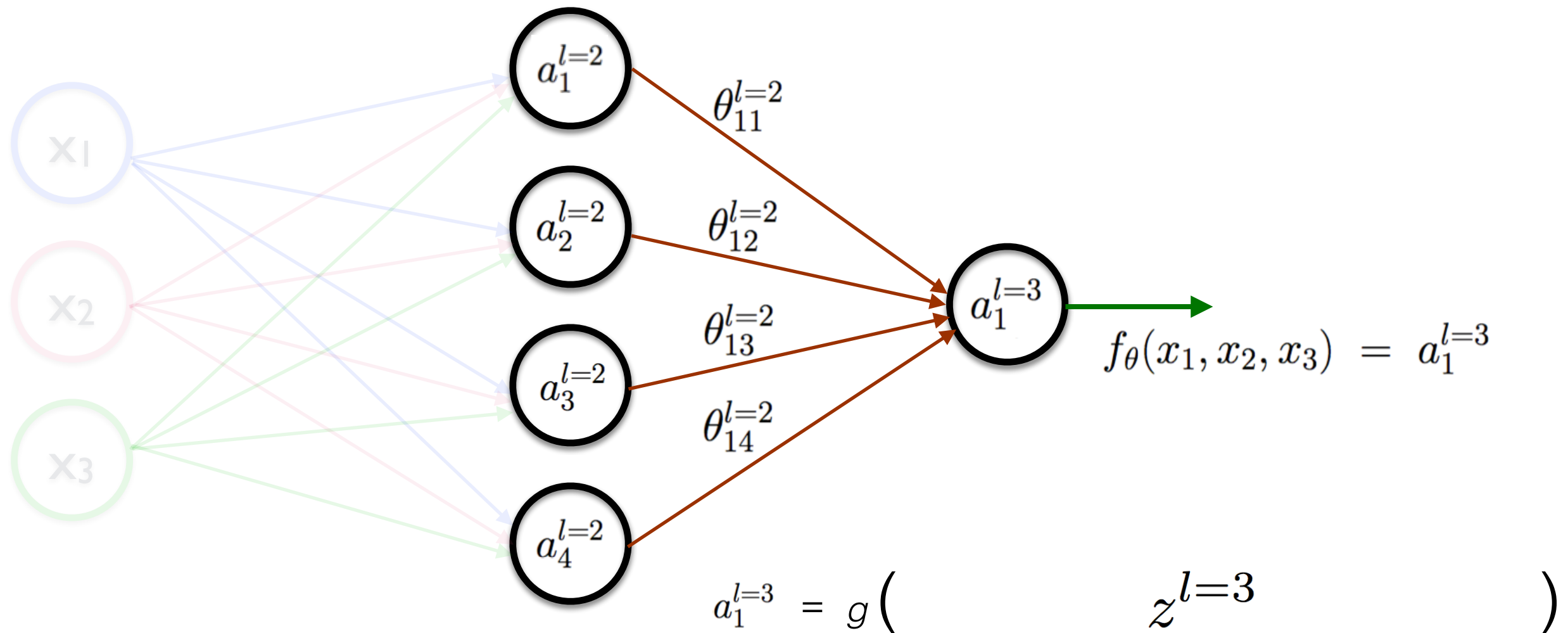
Vectorization



Vectorization

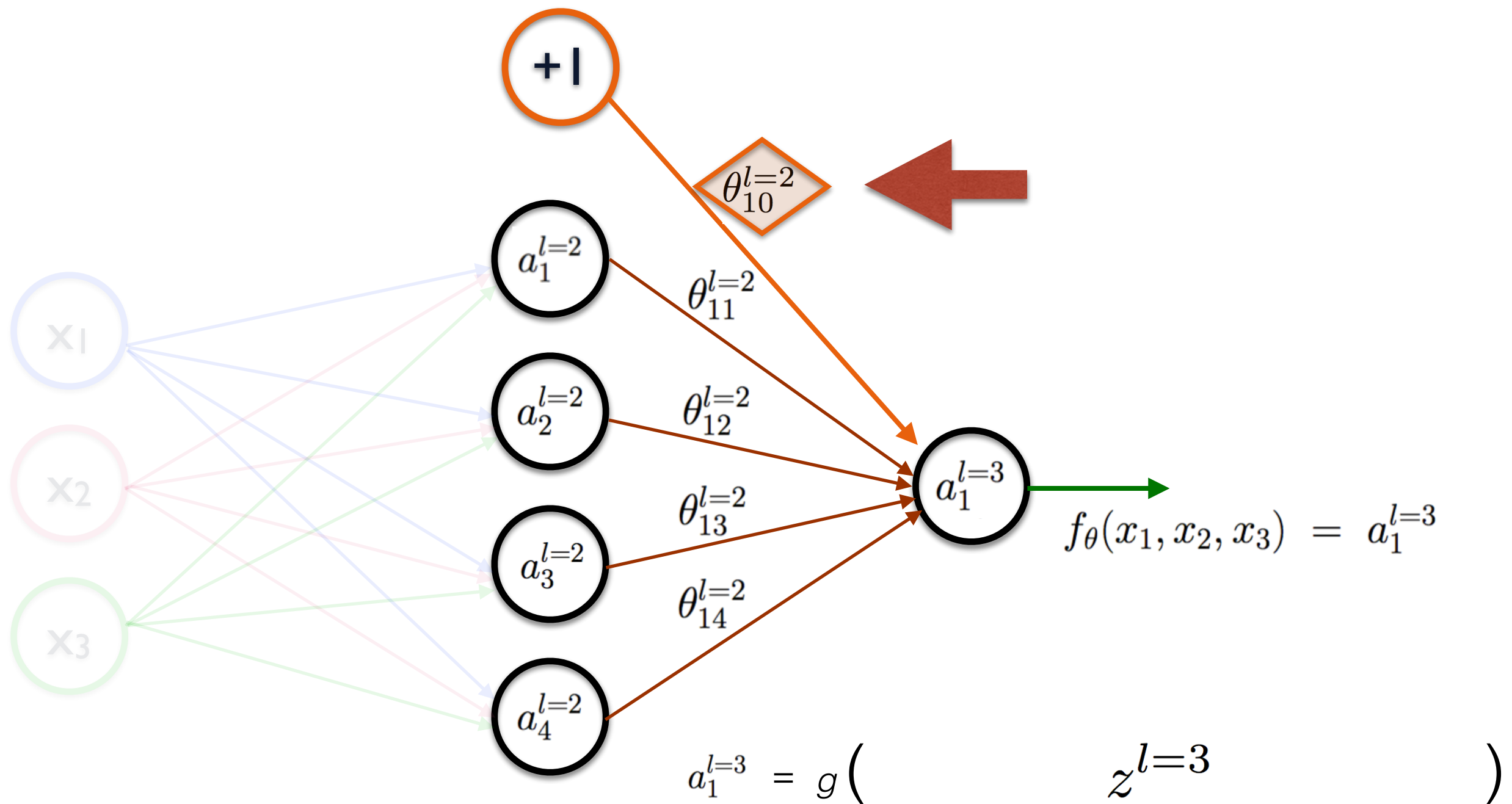


What about the *bias* term?



$$\begin{bmatrix} \theta_{11}^{l=2} & \theta_{12}^{l=2} & \theta_{13}^{l=2} & \theta_{14}^{l=2} \end{bmatrix} \mathbf{x} \begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

$\theta^{l=2} \in \mathbb{R}^{1 \times 4}$
 $a^{l=2} \in \mathbb{R}^{4 \times 1}$



$$\begin{bmatrix} \theta_{11}^{l=2} & \theta_{12}^{l=2} & \theta_{13}^{l=2} & \theta_{14}^{l=2} \end{bmatrix} \mathbf{x} \begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

$\theta^{l=2} \in \mathbb{R}^{1 \times 4}$
 $a^{l=2} \in \mathbb{R}^{4 \times 1}$

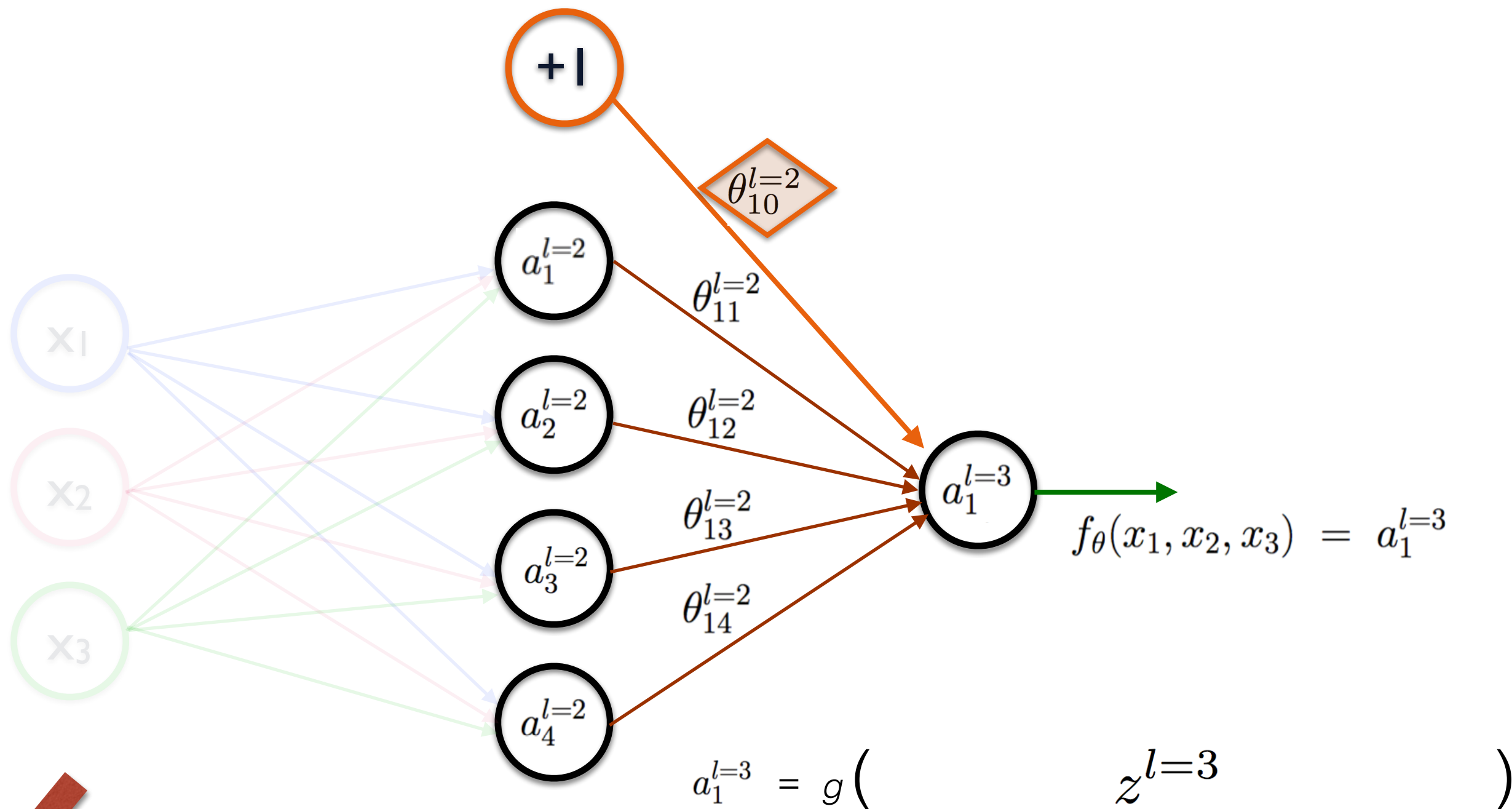


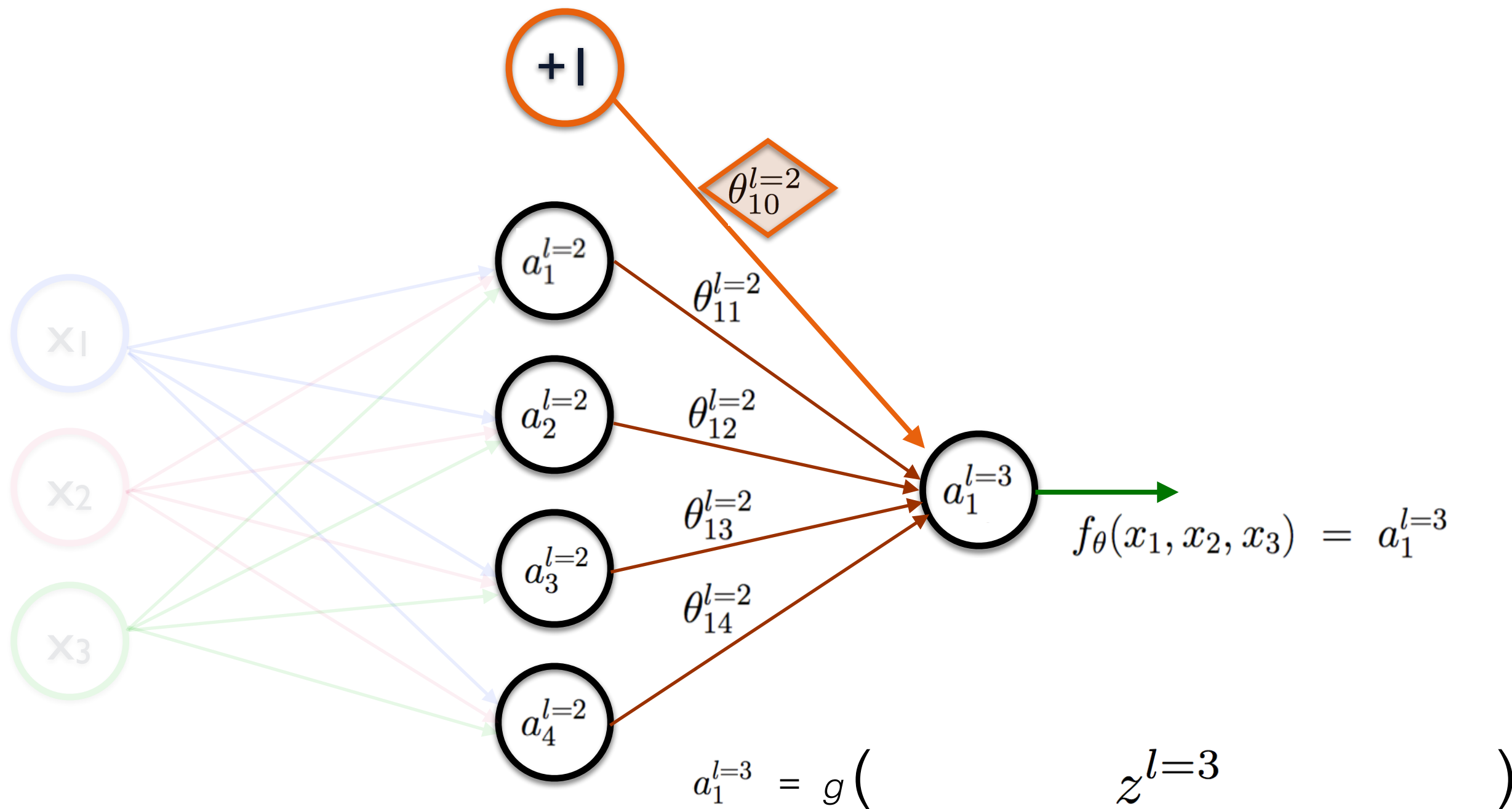
Diagram illustrating the matrix representation of the layer transition from layer $l=2$ to layer $l=3$.

The weight matrix $\theta^{l=2} \in \mathbb{R}^{1 \times 4}$ is shown as a row vector of weights $\theta_{10}^{l=2}, \theta_{11}^{l=2}, \theta_{12}^{l=2}, \theta_{13}^{l=2}, \theta_{14}^{l=2}$.

The hidden node vector $a^{l=2} \in \mathbb{R}^{4 \times 1}$ is shown as a column vector of hidden node values $a_1^{l=2}, a_2^{l=2}, a_3^{l=2}, a_4^{l=2}$.

The output node value $z^{l=3} \in \mathbb{R}^{1 \times 1}$ is calculated as:

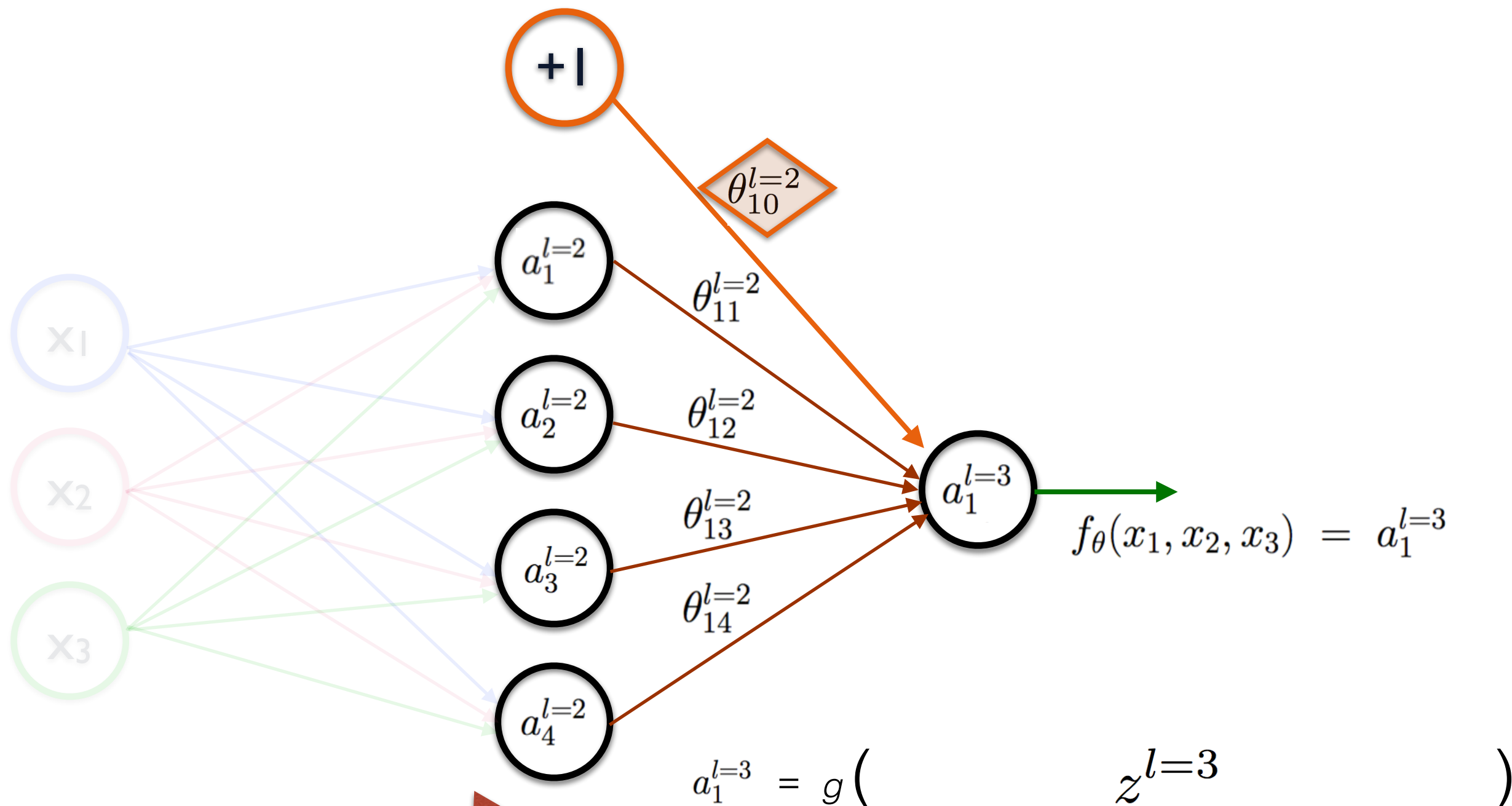
$$\theta^{l=2} \times a^{l=2} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$



$$[\theta_{10}^{l=2} \quad \theta_{11}^{l=2} \quad \theta_{12}^{l=2} \quad \theta_{13}^{l=2} \quad \theta_{14}^{l=2}] \mathbf{x} \begin{bmatrix} a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

$\theta^{l=2} \in \mathbb{R}^{1 \times 5}$

$a^{l=2} \in \mathbb{R}^{4 \times 1}$

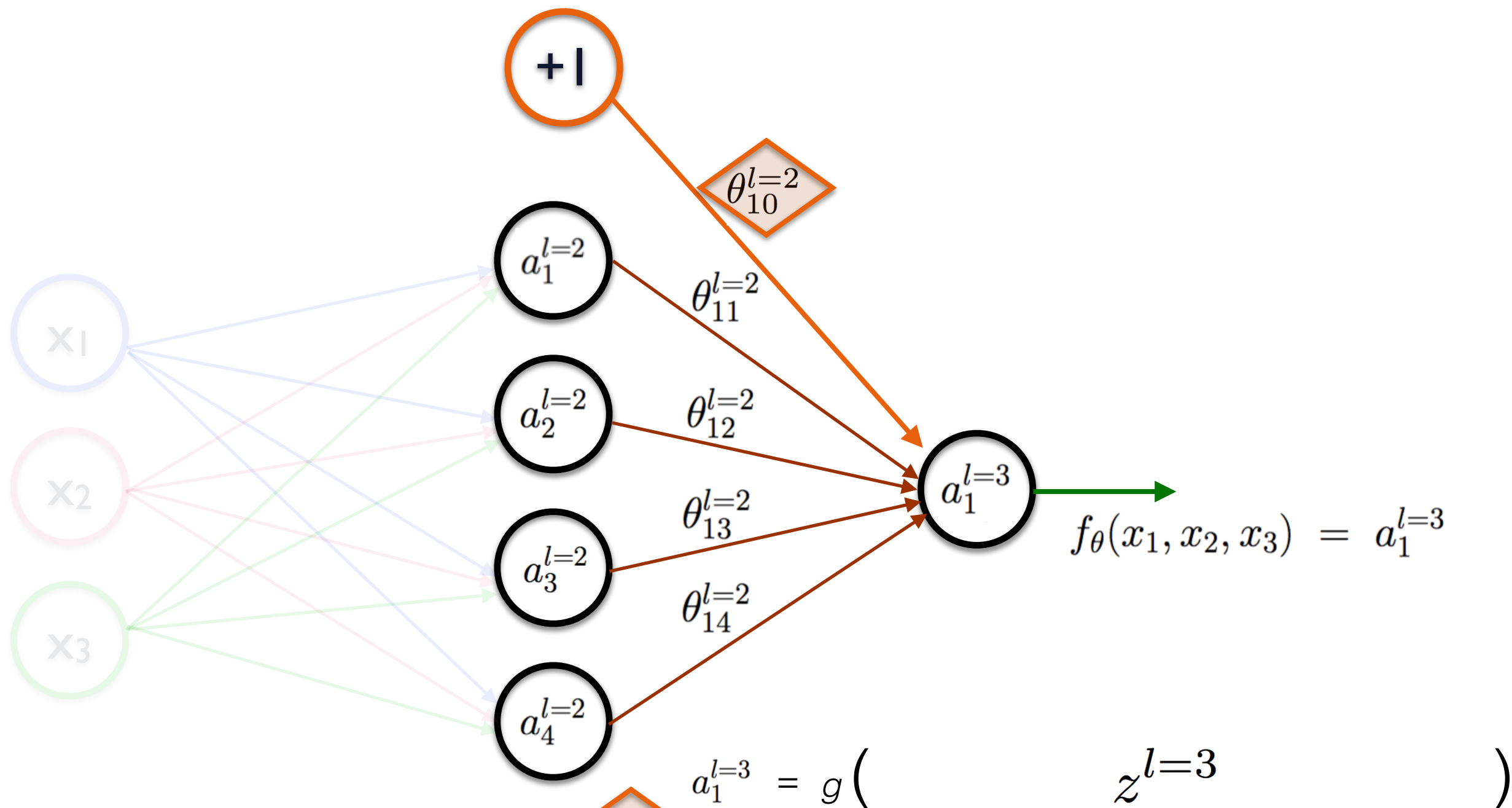


$$a_1^{l=3} = g \left(z^{l=3} \right)$$

$$\begin{bmatrix} \theta_{10}^{l=2} & \theta_{11}^{l=2} & \theta_{12}^{l=2} & \theta_{13}^{l=2} & \theta_{14}^{l=2} \end{bmatrix} \mathbf{x} \begin{bmatrix} 1 \\ a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

$$\theta^{l=2} \in \mathbb{R}^{1 \times 5}$$

$$a^{l=2} \in \mathbb{R}^{4 \times 1}$$



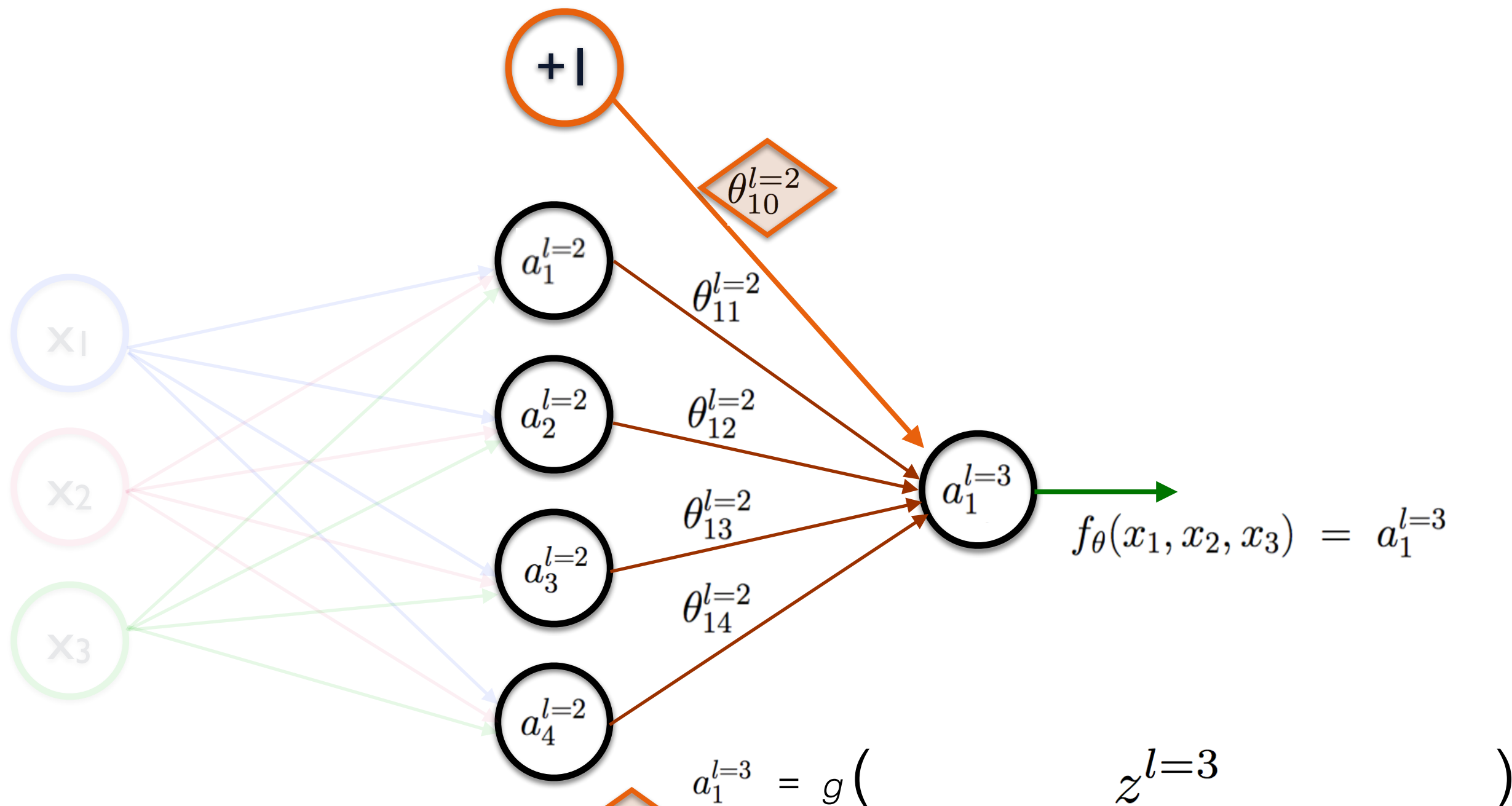
$$a_1^{l=3} = g \left(z^{l=3} \right)$$

$$\begin{bmatrix} \theta_{10}^{l=2} & \theta_{11}^{l=2} & \theta_{12}^{l=2} & \theta_{13}^{l=2} & \theta_{14}^{l=2} \end{bmatrix} \mathbf{x} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

$$\theta^{l=2} \in \mathbb{R}^{1 \times 5}$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix}$$

$$a^{l=2} \in \mathbb{R}^{4 \times 1}$$



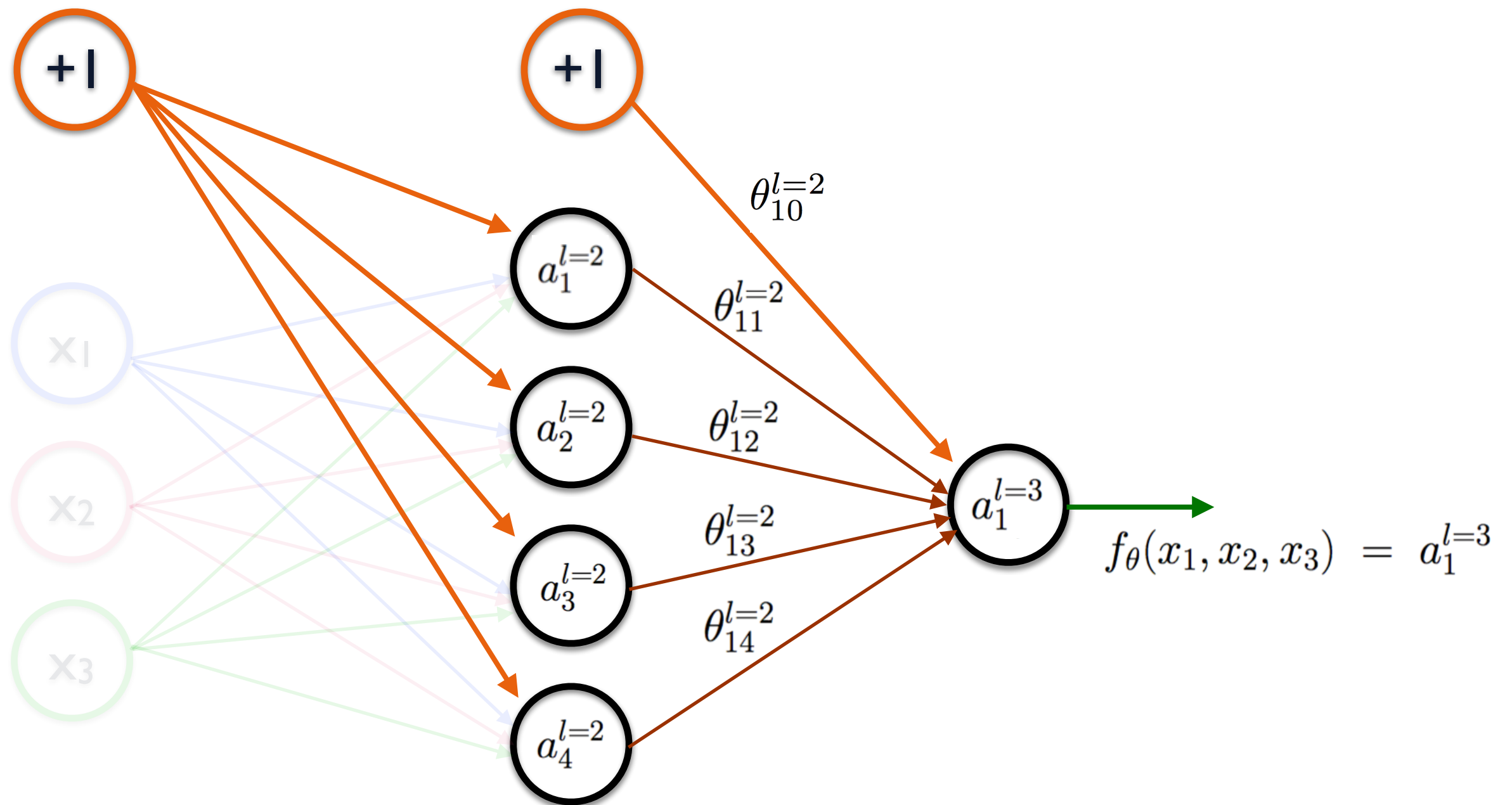
$$a_1^{l=3} = g \left(z^{l=3} \right)$$

$$\begin{bmatrix} \theta_{10}^{l=2} & \theta_{11}^{l=2} & \theta_{12}^{l=2} & \theta_{13}^{l=2} & \theta_{14}^{l=2} \end{bmatrix} \mathbf{x} = z^{l=3} \in \mathbb{R}^{1 \times 1}$$

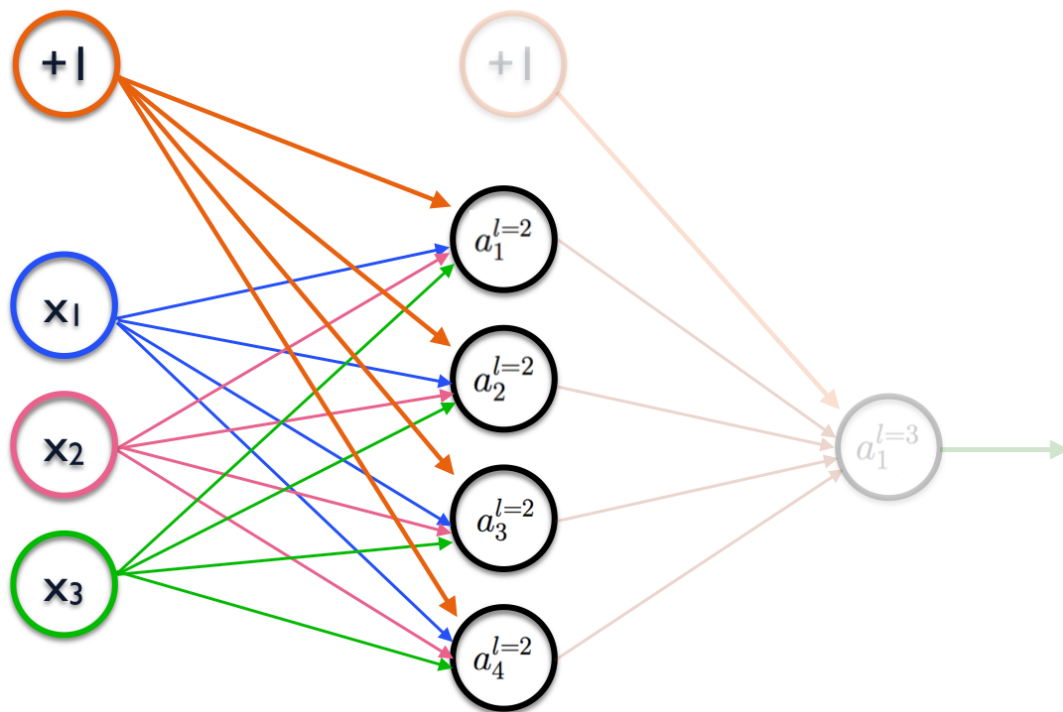
$$\theta^{l=2} \in \mathbb{R}^{1 \times 5}$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ a_1^{l=2} \\ a_2^{l=2} \\ a_3^{l=2} \\ a_4^{l=2} \end{bmatrix}$$

$$a^{l=2} \in \mathbb{R}^{5 \times 1}$$



Before introducing the bias term



$$a_1^{l=2} = g\left($$

$$a_2^{l=2} = g\left($$

$$a_3^{l=2} = g\left($$

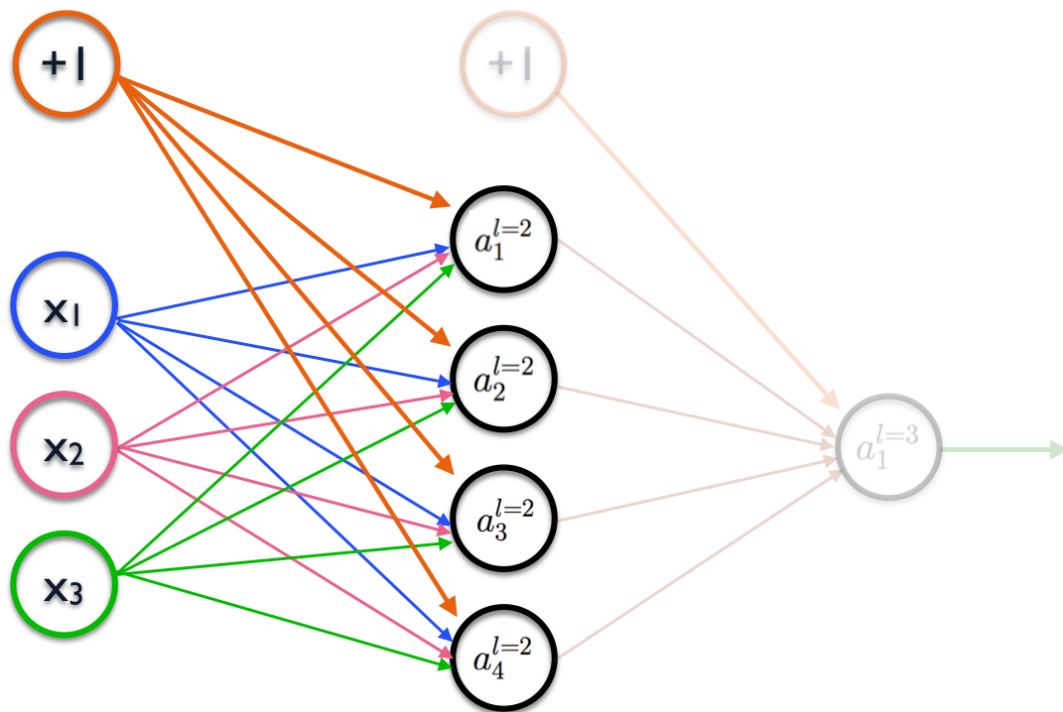
$$a_4^{l=2} = g\left($$

$$\begin{aligned} & x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1} \\ & x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1} \\ & x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1} \\ & x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1} \end{aligned}$$

$$\begin{bmatrix} \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \mathbf{x} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} =$$

$$\theta^{l=1} \mathbf{x} x^{(i)} = z^{l=2}$$

After introducing the bias term

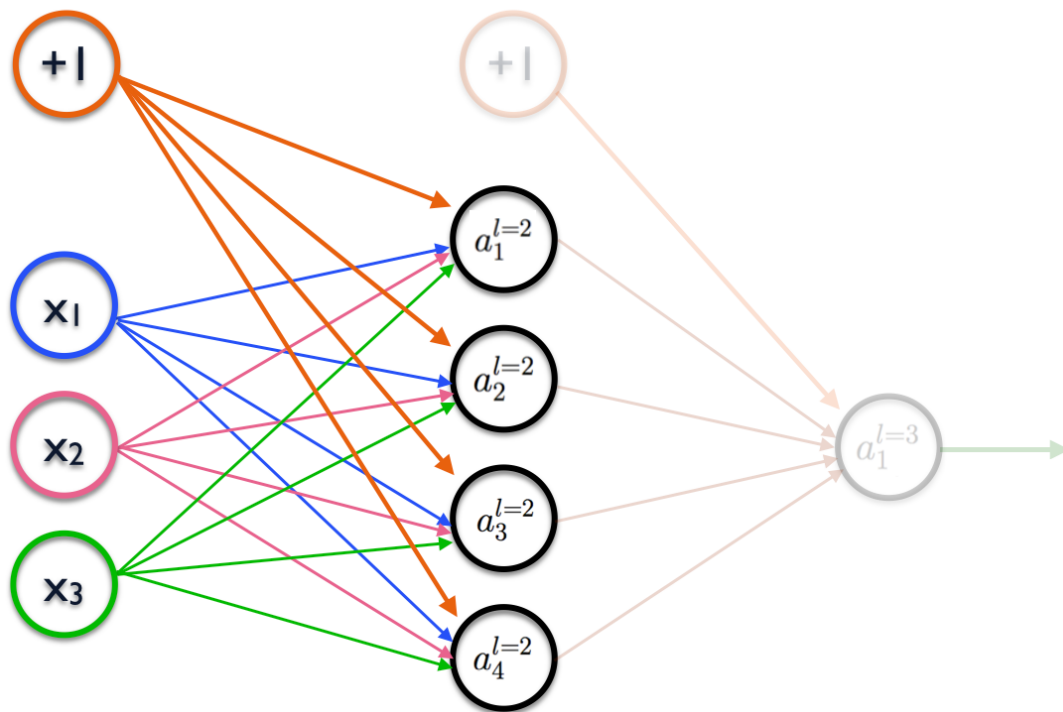


$$\begin{aligned}
 a_1^{l=2} &= g\left(\theta_{10}^{l=1} + x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(\theta_{20}^{l=1} + x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(\theta_{30}^{l=1} + x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(\theta_{40}^{l=1} + x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{10}^{l=1} & \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{20}^{l=1} & \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{30}^{l=1} & \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{40}^{l=1} & \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \mathbf{x} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{z}^{l=2}$$

$\theta^{l=1} \quad \mathbf{x} \quad x^{(i)} \quad = \quad z^{l=2}$

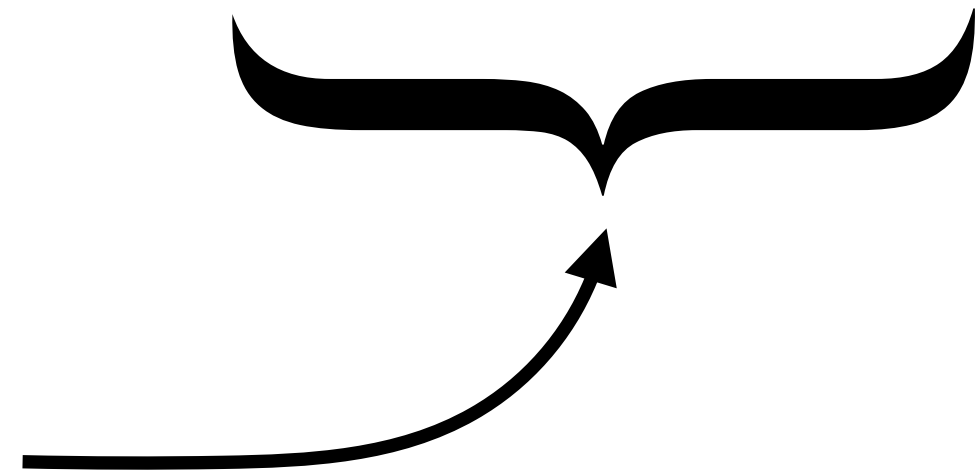
After introducing the bias term



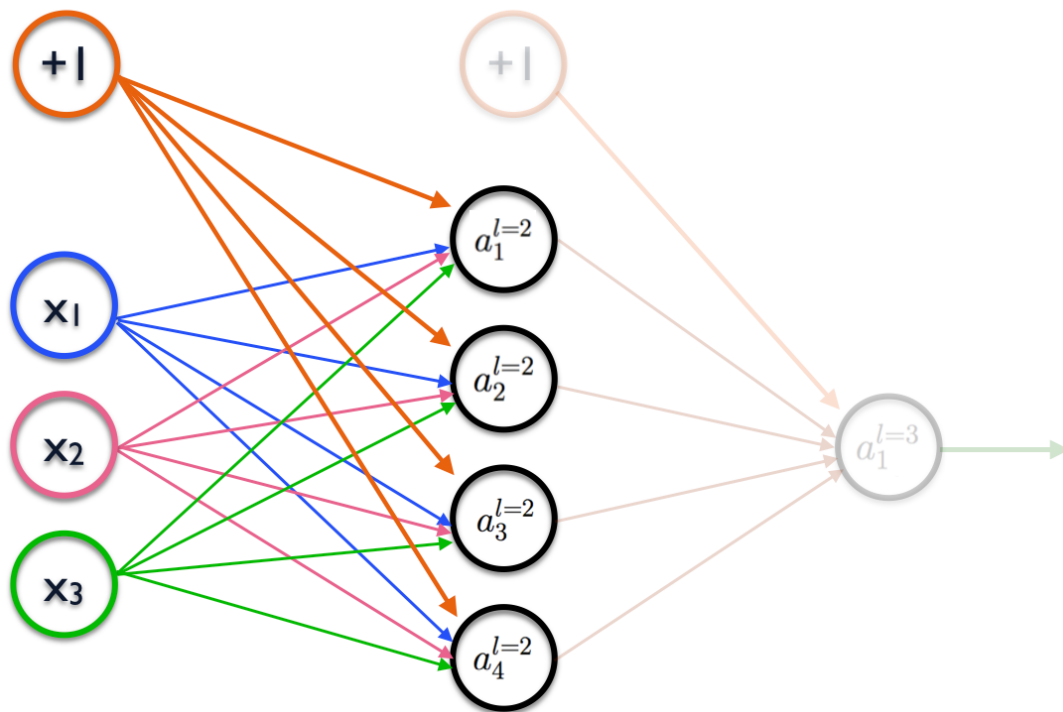
$$\begin{aligned}
 a_1^{l=2} &= g\left(\theta_{10}^{l=1} + x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(\theta_{20}^{l=1} + x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(\theta_{30}^{l=1} + x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(\theta_{40}^{l=1} + x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{10}^{l=1} & \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{20}^{l=1} & \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{30}^{l=1} & \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{40}^{l=1} & \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \mathbf{x} \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} =$$

$$\theta^{l=1} \mathbf{x} x^{(i)} = z^{l=2}$$



After introducing the bias term

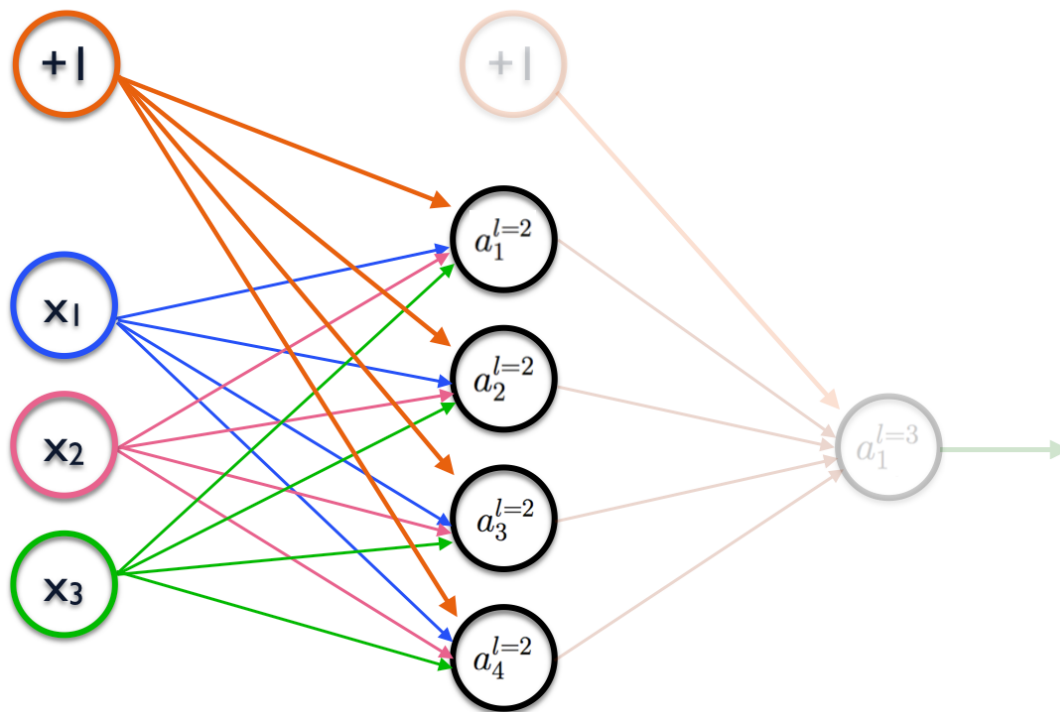


$$\begin{aligned}
 a_1^{l=2} &= g\left(\theta_{10}^{l=1} + x_1 \theta_{11}^{l=1} + x_2 \theta_{12}^{l=1} + x_3 \theta_{13}^{l=1}\right) \\
 a_2^{l=2} &= g\left(\theta_{20}^{l=1} + x_1 \theta_{21}^{l=1} + x_2 \theta_{22}^{l=1} + x_3 \theta_{23}^{l=1}\right) \\
 a_3^{l=2} &= g\left(\theta_{30}^{l=1} + x_1 \theta_{31}^{l=1} + x_2 \theta_{32}^{l=1} + x_3 \theta_{33}^{l=1}\right) \\
 a_4^{l=2} &= g\left(\theta_{40}^{l=1} + x_1 \theta_{41}^{l=1} + x_2 \theta_{42}^{l=1} + x_3 \theta_{43}^{l=1}\right)
 \end{aligned}$$

$$\begin{bmatrix} \theta_{10}^{l=1} & \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{20}^{l=1} & \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{30}^{l=1} & \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{40}^{l=1} & \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \mathbf{x} \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = z^{l=2}$$

$\theta^{l=1} \quad \mathbf{x} \quad a^{l=1} \quad = \quad z^{l=2}$

After introducing the bias term



$n_{l=1}$: number of neurons in layer $l=1$ (input)
 $n_{l=2}$: number of neurons in layer $l=2$ (hidden)
 $n_{l=3}$: number of neurons in layer $l=3$ (output)

$$\begin{bmatrix} \theta_{10}^{l=1} & \theta_{11}^{l=1} & \theta_{12}^{l=1} & \theta_{13}^{l=1} \\ \theta_{20}^{l=1} & \theta_{21}^{l=1} & \theta_{22}^{l=1} & \theta_{23}^{l=1} \\ \theta_{30}^{l=1} & \theta_{31}^{l=1} & \theta_{32}^{l=1} & \theta_{33}^{l=1} \\ \theta_{40}^{l=1} & \theta_{41}^{l=1} & \theta_{42}^{l=1} & \theta_{43}^{l=1} \end{bmatrix} \mathbf{x} \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = z^{l=2}$$

$$\mathbb{R}^{n_{l=2} \times (n_{l=1} + 1)} \mathbf{x} a^{l=1}_{\mathbb{R}^{(n_{l=1} + 1) \times 1}} = z^{l=2}_{\mathbb{R}^{n_{l=2} \times 1}}$$

Forward Propagation of One Instance

L : number of layers

$n_{l=1}$: number of neurons in layer $l=1$ (input)

$n_{l=2}$: number of neurons in layer $l=2$ (hidden)

$\vdots$

$n_{l=L}$: number of neurons in layer $l=L$ (output)

$$\theta^{l=1} \in \mathbb{R}^{n_{l=2} \times (n_{l=1} + 1)}$$

$$\theta^{l=2} \in \mathbb{R}^{n_{l=3} \times (n_{l=2} + 1)}$$

$\vdots$

$$\theta^{l=L-1} \in \mathbb{R}^{n_{l=L} \times (n_{l=L-1} + 1)}$$

Forward Propagation of One Instance

L : number of layers

$n_{l=1}$: number of neurons in layer $l=1$ (input)

$n_{l=2}$: number of neurons in layer $l=2$ (hidden)

$\vdots$

$n_{l=L}$: number of neurons in layer $l=L$ (output)

$$\theta^{l=1} \in \mathbb{R}^{n_{l=2} \times (n_{l=1} + 1)}$$

$$\theta^{l=2} \in \mathbb{R}^{n_{l=3} \times (n_{l=2} + 1)}$$

$\vdots$

$$\theta^{l=L-1} \in \mathbb{R}^{n_{l=L} \times (n_{l=L-1} + 1)}$$

1. For a given training instance $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$:

1. $\mathbf{a}^{l=1} = \mathbf{x}^{(i)}$

2. Add a bias term to $\mathbf{a}^{l=1}$ // prepares the network input

3. For each layer $k=2 \dots L-1$:

1. $\mathbf{z}^{(l=k)} = \boldsymbol{\theta}^{(l=k-1)} \mathbf{a}^{(l=k-1)}$

2. $\mathbf{a}^{(l=k)} = \mathbf{g}(\mathbf{z}^{(l=k)})$ // activation/output of all neurons in the k -th hidden layer

3. Adds bias term to $\mathbf{a}^{(l=k)}$

4. $\mathbf{z}^{(l=L)} = \boldsymbol{\theta}^{(l=L-1)} \mathbf{a}^{(l=L-1)}$ // activation/output of all neurons in the output layer

5. $\mathbf{a}^{(l=L)} = \mathbf{g}(\mathbf{z}^{(l=L)})$ // activation/output of all neurons in the output layer

6. Return $\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) = \mathbf{a}^{(l=L)}$

A brief comment on notation

Matrix/vector multiplication

E.g., matrix times matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} * \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

E.g., vector times vector

$$\begin{bmatrix} a \\ b \end{bmatrix}^T * \begin{bmatrix} c \\ d \end{bmatrix} = [a \quad b] * \begin{bmatrix} c \\ d \end{bmatrix} = [ac + bd]$$

A brief comment on notation

Element-wise multiplication

E.g., matrix times matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot * \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae & bf \\ cg & dh \end{bmatrix}$$

E.g., vector times vector

$$\begin{bmatrix} a \\ b \end{bmatrix} \cdot * \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac \\ bd \end{bmatrix}$$

L : number of layers

$n_{l=1}$: number of neurons in layer $l=1$ (input)

$n_{l=2}$: number of neurons in layer $l=2$ (hidden)

$\vdots$

$n_{l=L}$: number of neurons in layer $l=L$ (output)

$$\theta^{l=1} \in \mathbb{R}^{n_{l=2} \times (n_{l=1} + 1)}$$

$$\theta^{l=2} \in \mathbb{R}^{n_{l=3} \times (n_{l=2} + 1)}$$

$\vdots$

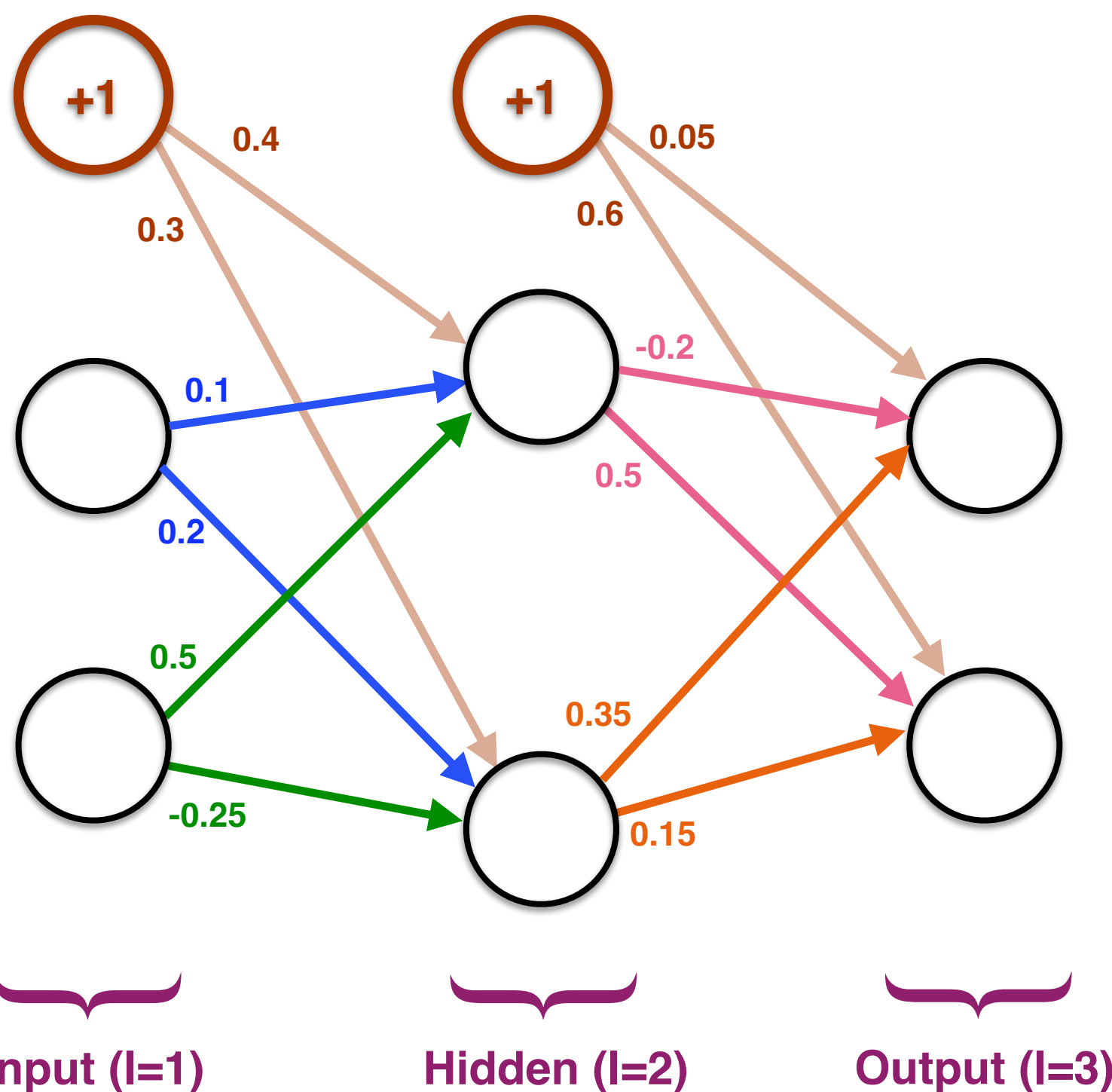
$$\theta^{l=L-1} \in \mathbb{R}^{n_{l=L} \times (n_{l=L-1} + 1)}$$

Demo

Forward Propagation

Demo

Forward Propagation



Training set:

$$\left. \begin{array}{ll} x_1^{(1)} = 0.13 & y_1^{(1)} = 0.7 \\ x_2^{(1)} = 0.41 & y_2^{(1)} = 0.3 \end{array} \right\} \text{Instance } ((x^{(1)}, y^{(1)}))$$

$$\left. \begin{array}{ll} x_1^{(2)} = 0.21 & y_1^{(2)} = 0.4 \\ x_2^{(2)} = 0.17 & y_2^{(2)} = 0.6 \end{array} \right\} \text{Instance } ((x^{(2)}, y^{(2)}))$$

Assume $\lambda = 0.93$

$$f(x^{(1)}) = \begin{bmatrix} 0.5286 \\ 0.7327 \end{bmatrix} \quad f(x^{(2)}) = \begin{bmatrix} 0.5315 \\ 0.7307 \end{bmatrix}$$

Cost Function J

L : number of layers

$n_{l=1}$: number of neurons in layer $l=1$ (input)

$n_{l=2}$: number of neurons in layer $l=2$ (hidden)

$\vdots$

$n_{l=L}$: number of neurons in layer $l=L$ (output)

$$\theta^{l=1} \in \mathbb{R}^{n_{l=2} \times (n_{l=1} + 1)}$$

$$\theta^{l=2} \in \mathbb{R}^{n_{l=3} \times (n_{l=2} + 1)}$$

$\vdots$

$$\theta^{l=L-1} \in \mathbb{R}^{n_{l=L} \times (n_{l=L-1} + 1)}$$

Cost Function J

L: number of layers

$n_{l=1}$: number of neurons in layer $l=1$ (input)

$n_{l=2}$: number of neurons in layer $l=2$ (hidden)

$\vdots$

$n_{l=L}$: number of neurons in layer $l=L$ (output)

$$\theta^{l=1} \in \mathbb{R}^{n_{l=2} \times (n_{l=1} + 1)}$$

$$\theta^{l=2} \in \mathbb{R}^{n_{l=3} \times (n_{l=2} + 1)}$$

$\vdots$

$$\theta^{l=L-1} \in \mathbb{R}^{n_{l=L} \times (n_{l=L-1} + 1)}$$

1. **J=0** // initializes the variable that will accumulate the total cost/error incurred by the network
2. For each training instance **($\mathbf{x}^{(i)}$, $\mathbf{y}^{(i)}$)**:
 1. Propagate $\mathbf{x}^{(i)}$ and compute each of the network's outputs, $\mathbf{f}_{\theta}(\mathbf{x}^{(i)})$
 2. Compute a vector $\mathbf{J}^{(i)}$ containing the cost/error associated with each of the network's outputs:

$$\mathbf{J}^{(i)} = -\mathbf{y}^{(i)} .* \log(\mathbf{f}_{\theta}(\mathbf{x}^{(i)})) - (1-\mathbf{y}^{(i)}) .* \log(1 - \mathbf{f}_{\theta}(\mathbf{x}^{(i)}))$$

$$\mathbf{J} = \mathbf{J} + \text{sum}(\mathbf{J}^{(i)})$$
 // adds all elements of the vector $\mathbf{J}^{(i)}$ and accumulates this sum in J
3. **J = J / n** // divides the total error/cost of the network by the number of training instances
4. **S** = computes the square of all weights of the network (except bias weights) and adds them up
5. **S = (λ/(2n)) * S** // computes the term used to regularize the network's cost
6. Return the regularized error/cost of the network, **J+S**, with respect to the entire training set

L : number of layers

$n_{l=1}$: number of neurons in layer $l=1$ (input)

$n_{l=2}$: number of neurons in layer $l=2$ (hidden)

$\vdots$

$n_{l=L}$: number of neurons in layer $l=L$ (output)

$$\theta^{l=1} \in \mathbb{R}^{n_{l=2} \times (n_{l=1} + 1)}$$

$$\theta^{l=2} \in \mathbb{R}^{n_{l=3} \times (n_{l=2} + 1)}$$

$\vdots$

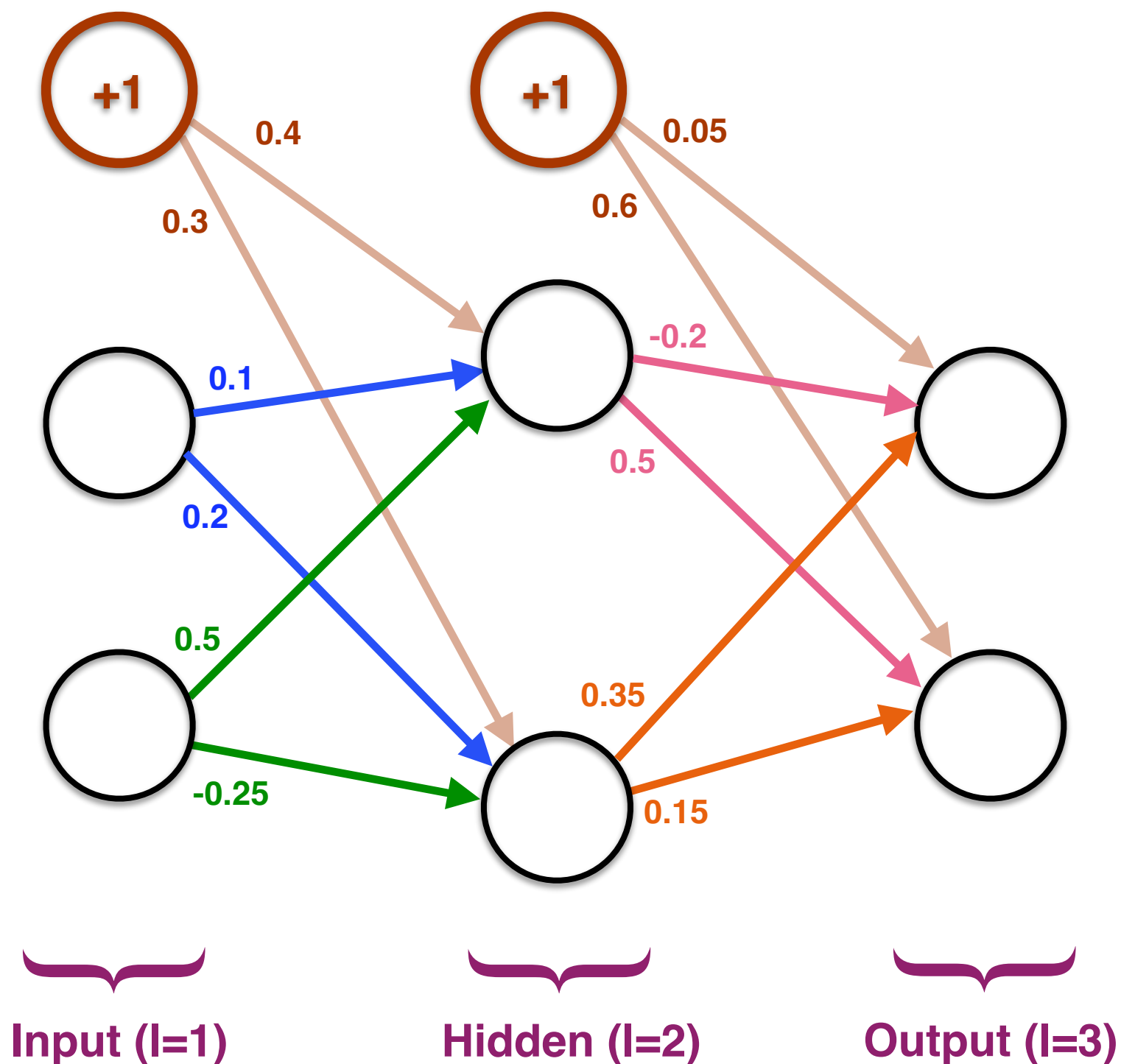
$$\theta^{l=L-1} \in \mathbb{R}^{n_{l=L} \times (n_{l=L-1} + 1)}$$

Demo

Forward Propagation + Cost Function

Demo

Forward Propagation + Cost Function



Training set:

$x_1^{(1)} = 0.13$	$y_1^{(1)} = 0.7$	} Instance $((x^{(1)}, y^{(1)}))$
$x_2^{(1)} = 0.41$	$y_2^{(1)} = 0.3$	
$x_1^{(2)} = 0.21$	$y_1^{(2)} = 0.4$	} Instance $((x^{(2)}, y^{(2)}))$
$x_2^{(2)} = 0.17$	$y_2^{(2)} = 0.6$	

Assume $\lambda = 0.93$

$$f(x^{(1)}) = \begin{bmatrix} 0.5286 \\ 0.7327 \end{bmatrix} \quad f(x^{(2)}) = \begin{bmatrix} 0.5315 \\ 0.7307 \end{bmatrix}$$

$$J^{(1)} = \begin{bmatrix} 0.6719 \\ 1.0168 \end{bmatrix} \quad J^{(2)} = \begin{bmatrix} 0.7078 \\ 0.7130 \end{bmatrix}$$

$$J = 1.5547 \quad S = 0.1854$$

$$\text{Regularized cost: } J + S = 1.7402$$

Backpropagation

1. For each training instance $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$:

1.1. Propagate $\mathbf{x}^{(i)}$ and compute each of the network's outputs, $\mathbf{f}_{\theta}(\mathbf{x}^{(i)})$

1.2. $\delta^{(l=L)} = \mathbf{f}_{\theta}(\mathbf{x}^{(i)}) - \mathbf{y}^{(i)}$ // computes the delta values of all output neurons

1.3. For each network layer, $k=L-1 \dots 2$ // computes the delta values of all neurons in the hidden layers

$$\delta^{(l=k)} = [\theta^{(l=k)}]^T \delta^{(l=k+1)} .* \mathbf{a}^{(l=k)} .* (1 - \mathbf{a}^{(l=k)})$$

Removes the first element of $\delta^{(l=k)}$ (i.e., the delta value associated with the bias neuron of the k -th layer of the network)

1.4. For each network layer, $k=L-1 \dots 1$ // updates gradients of the weights of each layer, based on the current training instance

$$\mathbf{D}^{(l=k)} = \mathbf{D}^{(l=k)} + \delta^{(l=k+1)} [\mathbf{a}^{(l=k)}]^T$$
 // accumulates, in $\mathbf{D}^{(l=k)}$, gradients computed based on the current training instance

2. For each network layer, $k=L-1 \dots 1$ // computes the final (regularized) gradients of the weights of each layer

2.1. Let $\mathbf{P}^{(l=k)}$ be equal to $(\lambda .* \theta^{(l=k)})$, // computes regularizer, λ , of all non-bias weights and set its first column to be all zeros

2.2. $\mathbf{D}^{(l=k)} = (1/n) (\mathbf{D}^{(l=k)} + \mathbf{P}^{(l=k)})$ // combines gradients w/ regularization terms; divides by #instances to obtain average gradient

3. // At this point, $\mathbf{D}^{(l=1)}$ contains the gradients of the weights $\theta^{(l=1)}$; (...); and $\mathbf{D}^{(l=L-1)}$ contains the gradients of the weights $\theta^{(l=L-1)}$

4. For each network layer, $k=L-1 \dots 1$ // updates the weights of each layer based on their corresponding gradients

$$4.1. \theta^{(l=k)} = \theta^{(l=k)} - \alpha .* \mathbf{D}^{(l=k)}$$

Backpropagation

1. For each training instance $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$:

1.1. Propagate $\mathbf{x}^{(i)}$ and compute each of the network's outputs, $\mathbf{f}_{\theta}(\mathbf{x}^{(i)})$

1.2. $\delta^{(l=L)} = \mathbf{f}_{\theta}(\mathbf{x}^{(i)}) - \mathbf{y}^{(i)}$ // computes the delta values of all output neurons

1.3. For each network layer, $k=L-1 \dots 2$ // computes the delta values of all neurons in the hidden layers

$$\delta^{(l=k)} = [\boldsymbol{\theta}^{(l=k)}]^T \delta^{(l=k+1)} .* \mathbf{a}^{(l=k)} .* (1 - \mathbf{a}^{(l=k)})$$

Removes the first element of $\delta^{(l=k)}$ (i.e., the delta value associated with the bias neuron of the k -th layer of the network)

1.4. For each network layer, $k=L-1 \dots 1$ // updates gradients of the weights of each layer, based on the current training instance

$$\mathbf{D}^{(l=k)} = \mathbf{D}^{(l=k)} + \delta^{(l=k+1)} [\mathbf{a}^{(l=k)}]^T$$
 // accumulates, in $\mathbf{D}^{(l=k)}$, gradients computed based on the current training instance

2. For each network layer, $k=L-1 \dots 1$ // computes the final (regularized) gradients of the weights of each layer

2.1. Let $\mathbf{P}^{(l=k)}$ be equal to $(\boldsymbol{\lambda} .* \boldsymbol{\theta}^{(l=k)})$, // computes regularizer, $\boldsymbol{\lambda}$, of all non-bias weights and set its first column to be all zeros

2.2. $\mathbf{D}^{(l=k)} = (1/n) (\mathbf{D}^{(l=k)} + \mathbf{P}^{(l=k)})$ // combines gradients w/ regularization terms; divides by #instances to obtain average gradient

3. // At this point, $\mathbf{D}^{(l=1)}$ contains the gradients of the weights $\boldsymbol{\theta}^{(l=1)}$; (...); and $\mathbf{D}^{(l=L-1)}$ contains the gradients of the weights $\boldsymbol{\theta}^{(l=L-1)}$

4. For each network layer, $k=L-1 \dots 1$ // updates the weights of each layer based on their corresponding gradients

$$4.1. \boldsymbol{\theta}^{(l=k)} = \boldsymbol{\theta}^{(l=k)} - \alpha .* \mathbf{D}^{(l=k)}$$

Backpropagation

1. For each training instance $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$:
 - 1.1. Propagate $\mathbf{x}^{(i)}$ and compute each of the network's outputs, $\mathbf{f}_{\theta}(\mathbf{x}^{(i)})$
 - 1.2. $\delta^{(l=L)} = \mathbf{f}_{\theta}(\mathbf{x}^{(i)}) - \mathbf{y}^{(i)}$ // computes the delta values of all output neurons
 - 1.3. For each network layer, $k=L-1 \dots 2$ // computes the delta values of all neurons in the hidden layers
$$\delta^{(l=k)} = [\boldsymbol{\theta}^{(l=k)}]^T \delta^{(l=k+1)} \odot \mathbf{a}^{(l=k)} \odot (1 - \mathbf{a}^{(l=k)})$$

Removes the first element of $\delta^{(l=k)}$ (i.e., the delta value associated with the bias neuron of the k -th layer of the network)
 - 1.4. For each network layer, $k=L-1 \dots 1$ // updates gradients of the weights of each layer, based on the current training instance
$$\mathbf{D}^{(l=k)} = \mathbf{D}^{(l=k)} + \delta^{(l=k+1)} [\mathbf{a}^{(l=k)}]^T$$
 // accumulates, in $\mathbf{D}^{(l=k)}$, gradients computed based on the current training instance
2. For each network layer, $k=L-1 \dots 1$ // computes the final (regularized) gradients of the weights of each layer
 - 2.1. Let $\mathbf{P}^{(l=k)}$ be equal to $(\lambda \odot \boldsymbol{\theta}^{(l=k)})$, // computes regularizer, λ , of all non-bias weights and set its first column to be all zeros
 - 2.2. $\mathbf{D}^{(l=k)} = (1/n) (\mathbf{D}^{(l=k)} + \mathbf{P}^{(l=k)})$ // combines gradients w/ regularization terms; divides by #instances to obtain average gradient
3. // At this point, $\mathbf{D}^{(l=1)}$ contains the gradients of the weights $\boldsymbol{\theta}^{(l=1)}$; (...); and $\mathbf{D}^{(l=L-1)}$ contains the gradients of the weights $\boldsymbol{\theta}^{(l=L-1)}$
4. For each network layer, $k=L-1 \dots 1$ // updates the weights of each layer based on their corresponding gradients
 - 4.1. $\boldsymbol{\theta}^{(l=k)} = \boldsymbol{\theta}^{(l=k)} - \alpha \odot \mathbf{D}^{(l=k)}$

Repeat until stopping criterion is met



Machine Learning

CMPSCI 589

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Neural Networks (pt. 3)

Vectorization

